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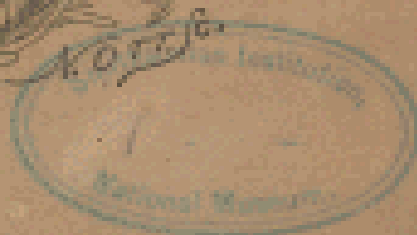
DAGUERREIAN JOURNAL:

Devoted to the Daguerreian and Photogenic Arts.

Also, embracing the Sciences, Arts, and Literature.



L. J. Rogers



VOL. I.

MARCH 1, 1851.

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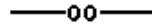


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ETCHING DAGUERRETYPE PLATES.

BY W. R. GROVE.

Dr. Berres of Vienna was the first, I believe, who published a process for etching Daguerreotypes; his method was to cover the plates with a solution of gum-arabic, and then to immerse them in nitric acid of a certain strength. I have not seen any plates thus prepared, but the few experiments which I have made with nitric acid, have given me a burred and imperfect outline; and I have experienced extreme difficulty of manipulation from the circumstance of the acid never attacking the plate uniformly and simultaneously. My object, however, in this communication, is not to find fault with a process which I have never perhaps fairly tried or seen tried by experienced hands, and the inventor of which deserves the gratitude of all interested in physical science; but to make public another which possesses the advantage of extreme simplicity, which any one, however unskilled in chemical manipulation, may practise with success, and which produces a perfect etching of the original image; so much so, that a plate thus etched can scarcely be distinguished from an actual Daguerreotype, preserving all the microscopic delicacy of the finest parts of the impression.

One sentence will convey the secret of this process; it is to make the Daguerreotype the anode^[A] of a voltaic combination, in a solution which will not of itself attack either silver or mercury, but of which, when electrolyzed, the anion will attack these metals unequally. This idea occurred to me soon after the publication of Daguerre's process; but, being then in the country, and unable to procure any plates, I allowed the matter to sleep; and other occupations prevented for some time any recurrence to it. Recently having heard much conversation as to the practicability or impracticability of Daguerreotype engraving, I became anxious to try a few experiments in pursuance of my original notion; and for this purpose applied in several quarters for Daguerreotypes; but, thanks to the exclusiveness of M. Daguerre's patent, I found that to procure a sufficient number of plates for any reasonable chance of success, was quite out of the question.

[A] Strictly speaking, this is a misapplication of Faraday's term; he applied it to the surface of the electrotpe; as, however, all continental, and many English writers (among whom I may name Whewell) have applied it to the positive electrode, and as an expression is most needed for that, I have not hesitated so to apply it.

On mentioning the subject to Mr. Gassiot, he, with his usual energy and liberality, offered to procure me a sufficiency of Daguerreotypes; and it is owing to his zealous

and valuable coöperation that I have been able to get such definite results as appear worth publication.

Five points naturally present themselves to the consideration of the experimenter on this subject: first, the quantity of the voltaic current; secondly, its intensity; thirdly, the distance between the anode and cathode; fourthly, the time during which the process should be continued; and fifthly, the solution to be employed.

1st. With regard to the first element or quantity, many previous experiments had convinced me that, to give the maximum and most uniform quantitative^[B] action of any voltaic combination, the electrodes should be of the same size as the generating plates; in other words, that the sectional area of the electrolyte should be the same throughout the whole voltaic circuit. It seems strange that this point should have been so generally overlooked as it has been; an electrician would never form a battery, one pair of plates of which were smaller than the rest; and yet the electrodes, which offering of themselves a resistance to the current, from the inoxidability of the anode, are, *a fortiori*, a restriction when of small size, have generally been formed indefinitely smaller than the generating plates; I, therefore, without further experiment, applied this principle to the process about to be detailed.

[B] I say the quantitative action; for where great intensity is required, as in decomposing alkalies, &c., it may be advisable to narrow the electrodes, so as to present a smaller surface for the reaction of the liberated elements.

2nd. *The intensity of the voltaic current.*—Here it appeared to me that, as in the electrolyte, where the visible action is at the cathode, a certain degree of intensity throws down metal as a crystal, an increased intensity as a metallic plate, and a further intensity as a pulverulent mass; that degree of intensity which would show on the negative deposit the finest impressions from the cathode, would also produce on the anode the most delicate excavations, and consequently, an intensity which would just fall short of the point of evolving oxygen from the plate to be etched, would be the most likely to succeed; this point was not, however, adopted without careful experiment, the more so, as in one instance Mr. Gassiot succeeded in procuring a very fair etching with a series of ten pairs of the nitric acid battery; however, the results of repeated experiments, in which the intensity has been varied from a series of sixteen pairs to one of the nitric acid battery, were strongly in favor of the above idea, and, consequently, went to prove that one pair gives the most efficient degree of intensity for the purpose required.

3rd. *The distance between the plates.*—As it was proved by De la Rive, that in an electrolytic solution, when the electrodes are at a distance, the action extends a little beyond the parallel lines which would join the bounds of the electrodes, and thus, that the current as it were diverges and converges, it appeared advisable to approximate the electrodes as nearly as possible, so as to produce uniformity of

action over the whole plate. Provided a solution be used which does not evolve gas at the cathode, I am inclined to think that the plates may be with advantage indefinitely approximated; but as this was not the case with the solution I selected for the greater number of experiments, 0·2 of an inch was fixed on as the distance, in order that the gas evolved from the cathode should not adhere to the anode, and thus interfere with the action.

4th. *Time of continuing the operation.*—This was a matter only to be decided by experiment, and must vary for the voltaic combination and solution employed. With a single pair of the nitric acid battery, from twenty-five to thirty seconds, was after a great number of experiments, fixed on as the proper time; and as the plate may at any period be removed from the solution and examined, the first experiment should never exceed twenty-five seconds, when, if not complete, the plate may be again subjected to electrolysis.

5th. *The solution to be employed.*—Here a vast field was open, and still is open to future experimentalists. Admitting the usual explanation of the Daguerreotype, which supposes the light parts to be mercury, and the dark silver, the object was to procure a solution which would attack one of these, and leave the other untouched. If one could be found to attack the silver and not the mercury, so much the better, as this would give a positive engraving, or one with lights and shadows, as in nature, while the converse would give a negative one. Unfortunately, silver and mercury are nearly allied in their electrical relations. I made several experiments with pure silver and mercury, used as the anode of a voltaic combination, but found that any solution which would act on one, acted also on the other. All then that could be expected, was a difference of action. With the Daguerreotype plates I have used the following:—

Dilute sulphuric acid, dilute hydrochloric acid, solution of sulphate of copper, of potash, and of acetate of lead. The object of using acetate of lead, was the following:— With this solution peroxide of lead is precipitated upon the anode; and, this substance being insoluble in nitric acid, it was hoped that the pure silver parts of the plate, being more closely invested with a stratum of peroxide than the mercurialized portions, these latter would, when immersed in this menstruum, be attacked, and thus furnish a negative etching. I was also not altogether without hopes of some curious effects, from the color of the thin films thus thrown down; here, however, I was disappointed; the colors succeeded each other as much as in the steel plate used for the metallochrome; but With inferior lustre. On immersion in nitric acid of different degrees of dilution, the plates were unequally attacked, and the etching burred and imperfect. Of the other solutions, hydrochloric acid was, after many experiments, fixed on as decidedly the best; indeed, this I expected, from the strong affinity of chlorine for silver.

I will now describe the manipulation which has been employed by Mr. Gassiot, and myself, in the laboratory of the London Institution, with very uniform success. A wooden frame is prepared, having two grooves at 0.2 of an inch distance, into which can be slid the plate to be etched, and a plate of platinum of the same size. To ensure a ready and equable evolution of hydrogen, this latter is platinized after Mr. Smee's method; for, if the hydrogen adhere to any part of the cathode, the opposite portions of the anode are proportionably less acted on. The back and edges of the Daguerreotype are varnished with a solution of shell-lac, which is scraped off one edge to admit of metallic connexion being established. The wooden frame with its two plates is now fitted into a vessel of glass or porcelain, filled with a solution of two measures hydrochloride acid, and one distilled water, (sp. gr. 1.1), and two stout platinum wires, proceeding from a single pair of the nitric acid battery, are made to touch the edges of the plates, while the assistant counts the time; this, as before stated, should not exceed thirty seconds. When the plate is removed from the acid, it should be well rinsed with distilled water; and will now (if the metal be homogeneous) present a beautiful sienna-colored drawing of the original design, produced by a film of the oxychloride formed;—it is then placed in an open dish containing a very weak solution of ammonia, and the surface gently rubbed with, very soft cotton, until all the deposit is dissolved; as soon as this is effected, it should be instantly removed, plunged into distilled water, and carefully dried. The process is now complete, and a perfect etching of the original design will be observed; this, when printed from, gives a positive picture, or one which has its lights and shadows as in nature; and which is, in this respect, more correct than the original Daguerreotype as the sides are not inverted; printing can therefore be directly read, and in portraits thus taken, the right and left sides of the face are in their proper position. There is, however, *ex necessitate rei*, this difficulty, with respect to prints from Daguerreotypes,—if the plates be etched to a depth sufficient to produce a very distinct impression, some of the finer lines of the original must inevitably run into each other, and thus the chief beauty of these exquisite images be destroyed. If, on the other hand, the process be only continued long enough to leave an exact etching of the original design, which can be done to the minutest perfection, the very cleaning of the plate by the printer destroys its beauty; and the molecules of the printing ink being larger than the depth of the etchings, an imperfect impression is produced. For this reason it appeared to me, that at present, the most important part of this process is the means it offers of multiplying indefinitely Daguerreotypes, by means of the electrotype. An ordinary Daguerreotype, it is known, will, when electrotyped, leave a faint impression; but in so doing it is entirely destroyed; and this impression cannot be perpetuated; but one thus etched at the voltaic anode, will admit of any number of copies being taken from it. To give an idea of the perfect accuracy of these, I may mention, that in one I have taken, on which is a sign-board measuring on the electrotype plate 0.1 by 0.06 of an inch, five lines of inscription

can, with the microscope, be distinctly read. The great advantages of the voltaic over the chemical process of etching, appear to me to be the following:—

1st. By the former, an indefinite variety of menstrua may be used; thus, solutions of acids, alkalies, salts, more especially the haloid class, sulphurets, cyanurets, in fact, any element which may be evolved by electrolysis, may be made to act upon the plate.

2nd. The action is generalized; and local voltaic currents are avoided.

3rd. The time of operation can be accurately determined; and any required depth of etching produced.

4th. The process can be stopped at any period, and again renewed if desirable.

The time I have given is calculated for experiments made with one pair of the nitric acid battery; it is, however, by no means necessary that this be employed, as probably any other form of voltaic combination may be efficient. It would seem more advisable to employ a diaphragm battery, or one which produces a constant current, as otherwise the time cannot be accurately determined. It is very necessary that the silver of plates subjected to this process be homogeneous. Striæ, imperceptible in the original Daguerreotype, are instantly brought out by the action of the nascent anion; probably silver, formed by voltaic precipitation, would be found the most advantageous. I transmit with this paper some specimens of the prints of the etched plates, and of electrotypes taken from them; and in conclusion would call attention to the remarkable instance which these offer, of the effects of the imponderable upon the ponderable: thus, instead of a plate being inscribed, as "drawn by Landseer, and engraved by Cousins," it would be "drawn by Light, and engraved by Electricity!"



A valuable receipt, which every Daguerreian should have in his memory, has been suggested from the fact that one of our fraternity lost his hand by an accidental cut, caused while fitting a likeness in a locket, by the points of copper cutting the palm of his hand, in such a manner as to render amputation necessary. *White of eggs is the best antidote against the poison of copper.*

For the Daguerreian Journal.

PAINTING—ITS ORIGIN AND HISTORY.

BY WILLIAM WALCUTT.

There is no doubt of painting being coeval with written language, or it may be said, that the simple representation of figures, on tablets prepared for the purpose, were the first means used by men to record particular events, actions, and so forth, and may therefore be dated back farther than any written language; and, indeed, may be considered the foundation from which it sprung, it being not improbable, because it is so natural, that the very first dwellers on the earth may have recorded their young history by means of representations of corporeal objects.

Many will say that such rude delineations were not painting, but everything has its beginning. From these simple lines has the perfection of painting sprung, what Artist does not remember the feeble attempts he made at representations when but a child; does he not remember them with pleasure, and did he not then, and does he not now, consider them, feeble as they were, paintings? The first efforts at delineation were painting, without regard to color, or without regard to their truth in delineation. The origin of what may now be called painting, as an art instructive or decorative, is altogether involved in obscurity—authorities differing widely on the subject. Pliny dates its invention to a period antecedent to the siege of Troy, and other writers after. The Egyptians boasted of possessing the art six thousand years before the Greeks, but this has always been doubted. They certainly practised painting at a very early period, and Cambyses destroyed all the monuments of art he found in Egypt, as much as was in his power, which is some argument in favor of the statement; and hence all such monuments as display specimens of the art, must be posterior to his invasion. The remains of Egyptian painting, now extant, are but poor representations of the human, and other figures, showing a meagre advance in the art, being weakly and falsely drawn with no natural grouping or attempts at sentiment. The Phœnicians were an early cultivated people, and have the credit of first inventing letters, and therefore may, by the best authority, lay claim to the invention of painting. King Solomon, one thousand years before Christ, held their ingenuity in such high regard, that he employed them as the master workmen in building his temple, and the scriptures describe the magnificence of the columns and their capitals, and the borders and cornishes and the doors, besides the great brazen sea, supported by twelve oxen. The coins also of this people, that have come down to us, show in design, and execution, a good condition in the arts.

The scriptures abound in references to the arts. When Jacob took away the daughter of Laban, she stole and secreted from search some of her father's household gods. One of the great commandments, is that the children of Israel shall "not make unto themselves any graven image, or any likeness of any thing that is in the heavens above, or in the earth beneath, or in the waters under the earth." As before mentioned, the temple of Solomon appears to have been richly and elaborately decorated with works of art; and the IV chapter Ezekiel commences with these words: "Thou also son of man, take thee a tile, and lay it before thee, and portray upon it the city, even Jerusalem." And in the XXIII chapter, 14 verse, it is written thus: "For when she saw men portrayed on the wall, the images of the Chaldeans portrayed with vermillion, girded with girdles about their loins, exceeding in dyed attire upon their heads." Also in the VII chapter, 10 verse: "So I went in and saw and beheld every form of creeping things, and abominable beasts, and all the idols of the house of Israel, portrayed upon the wall round about." Also in Numbers XXXIII chapter and 52 verse: "Then he shall drive out all the inhabitants of the land from before you, and destroy all their pictures, and destroy all their molten images."

The Persians, the Arabians, and the Parthians, from their peculiar education, have no claims whatever to the art of painting, a rude representation of the human figure, with them, was considered as vulgar as the appearance of a naked person. Their figures are almost invariably clothed in clumsy draperies, profusely painted. They were worshippers of fire, and used no representations of their deity; and it is a curious fact, that, in their successful invasion of Egypt, they carried away with them no taste for the arts, or imbibed any of the religious principles of that polytheistic country.

The art of painting appears to have remained in the same state in China, without any particular change from time immemorial. They have never attempted anything beyond mere imitation, and that entirely devoid of taste or truth. The human figure, with them, is a ludicrous deformity, and their perspective is gained, by piling one object above another, until the picture as all ground and no sky. Invention and Imagination were never known among them; and, although the elaborateness of many of their works is astonishing, yet with such a people, it would be useless to look for the origin and progress of art.

The arts of the Etruscans are famous for the vases they produced. Etruria, in its ancient state, was one of the most powerful and civilized countries in Italy. Though the history of this nation is involved in obscurity, as the Romans tried every means in their power to destroy all its claims to refinement, yet there are sufficient proofs remaining of the height and perfection to which they carried the fine arts. Near the town of Civita Vecchia stood the ancient Etruscan city of Tarquinia, near which are found numbers of sepulchral grottoes, many of which are decorated with paintings and figures much in the style of those on the Etruscan vases. Some of the pictures

represent combats, and others dances of females, executed with considerable spirit. The pottery before mentioned, however, affords the greatest number of their specimens of the art of design; the forms displayed in the contour of the vases, no less than the paintings with which they are decorated, show the wonderful attainment in elegance of design, purity of form, and ingenuity of delineation. The power over line, and the facility of execution they reached, may be easily conceived from the absorbent nature of the material upon which they wrought. No retouching was possible; but the whole must have been completely arranged in the mind of the artist before it could be struck off. Pliny states, that in his day, the town of Ardea, an ancient city of Etruria, contained some paintings which he ascribes to a period anterior to the founding of Rome, and mentions with surprise their then perfect state of preservation. At Lacurium also, he describes some pictures of Atalanta and Helen, which were simply painted on the wall, and exhibited great merit in execution. These Caligulo, after a fruitless attempt, failed in removing. Cere, another Etruscan city, boasted some paintings of an early date. All these specimens, although of remote date, have no positive clue by which to ascertain anything positive as to the origin of the art of painting. And we are obliged to turn to Greece for the foundation from whence sprung works still the admiration of the world.

Religion was the motive of Greek art; it was, therefore, natural that they should endeavor to invest their own authors, for they considered themselves of divine origin, with the most perfect forms; and as man possessed that exclusively, they completely and thoroughly studied the elements of his constitution. The climate was favorable to the development of that form, and the establishment of exercises by their civil and political institutions, created models in nature, which elevated Greek art to the highest excellence.

The next step of the art was the monogram, which is the outline of figures without light or shade, with the addition, however, of parts within the outline. From this the monochrom, or painting with a single color, and a white ground, then covered with punic wax, first amalgamated with a resinous pigment generally of a red, sometimes of a dark brown or black color, was the next advance. Through this inky ground, the outlines were traced with a firm though plyant style, called a cestrum: the line could be altered by the finger or a sponge, and easily replaced by a new one. When the whole was settled, it was suffered to dry, and covered with a brown encaustic varnish; the lights were worked over again, and rendered more brilliant with a more delicate point, according to the gradual advance from mere outline to some indication; and at last to masses of light and shade—thence to the superinduction of different colors, or the invention of polychrom, which, by the addition of the pencil to the style, raised the stained drawing to a legitimate picture, and at length produced that vaunted harmony—"the magic scale of Grecian color."

The period at which the pencil supplanted the cestrum cannot be ascertained. Apollodorus in the 93d Olympiad, and Zeuxis, in the 94th, are said to have used it with freedom and power. Parrhasius painted the battle of the Lapithæ and centaurs on the shield of Minerva for Phidias, to enable Mys to chase it. It was nearly a century after this that Appelles and Protogenes had a competition in drawing lines with the pencil, in which "delicacy and evanescent subtlety being the characteristic, some notion of their mechanical skill may be formed."

Encaustic painting was accomplished by using the colors in wax, as they are used now in oil, drying them by a fire, and polishing the surface by dry friction.

Polygnotus is the first great name that appears in history, that any satisfactory data of the arts may be commenced from. He lived about 400 years B. C. So great was his success in the Pœcile at Athens, and the Desche, or public hall at Delphi, that in a great council of the Amphyctons, it was solemnly decreed "that his expenses, whenever he travelled in Greece, should be borne at the public charge." His pictures were admired by Pliny, at the distance of six hundred years.

The first painting on record is the battle of Magnete, by Balarchus, and purchased by Candaules, King of Lydia, for its weight in gold, or, as some say, a quantity of gold coins equal to its surface.

After Aglaophon, Phidias, Panenus, Colotes, and Evenor, the father of Parrhasius, came Apollodorus, the Athenian. This painter applied the essential principles of Polygnotus to the delineation of the species, by investigating the leading forms that discriminate the different classes of human qualities and passions. The acuteness of his taste led him to discover, that, as all men were connected by one general form, so were they separated by some peculiar individuality. Pliny and Plutarch considered Apollodorus as the first colorist of his age, and it is very probable, by their descriptions, that he was the inventor of local color and tone. Zeuxis succeeded to Apollodorus, and by uniting in one figure the most perfect parts of many models, produced an ideal form, which, in his opinion, constituted the supreme degree of human beauty. Lucian describes a picture he exhibited at the Olympic Games as remarkable for its invention. It represented a female centaur, suckling her young. It was carried off from Athens by Sylla, but lost on the voyage to Italy.

Parrhasius, a native of Ephesus, but a citizen of Athens, was the son of a disciple of Evenor, and contemporary of Zeuxis. By his subtle examination of outline, "he established that standard of divine and heroic form, which raised him to the authority of a legislator, from whose decisions there was no appeal." He was a thorough master of allegory, from the fact of his embodying by signs, universally understood, the Athenian people. In a competition with Timanthes, he had the mortification of being

declared, by a majority of votes, inferior to him. The subject for competition, was the contest of Ajax and Ulysses for the arms of Achilles.

The sacrifice of Iphigenia in Aulis, by Timanthes, acquired the greatest celebrity of all the ancient pictures. Quintilian says that it was painted in contest with Colotes of Teos—an artist from the school of Phidias, and crowned with victory at its rival exhibition. This picture, which has been the subject of unlimited praise by the ancient critics, has been in modern times objected to, from the circumstance of Timanthes hiding the face of Agamemnon, the father of the victim, to be immolated in his mantle, unable, as it was supposed by his art, to express the entire agony of his grief. Sir Joshua Reynolds observes thus: "If difficulties overcome make a great part of the merit of art, difficulties evaded can deserve but little commendation." The French critic Falconet, has also condemned the artifice. But Fuseli answers these objections very reasonably by saying,—"The subject of Timanthes was the immolation of Sphigenia. Sphigenia was the principal figure, and her form, her resignation, or her anguish, was the painter's principal task; the figure of Agamemnon, however important, is merely accessory, and no more necessary to make the subject completely tragic, than that of Dytmemnestra, the mother—no more than that of Priam, to impress us with sympathy at the death of Polyxenia." Again, "they ascribe to impotence what was the forbearance of judgment." Timanthes felt like a father; he did not hide the face of Agamemnon, because it was beyond the power of his art—not because it was beyond the possibility, but because it was beyond the dignity of expression—because the inspiring feature of paternal affection at that moment, and the action which of necessity must have accompanied it, would either have destroyed the grandeur of the character, and the solemnity of the scene, or subject the painter, with the majority of his judges, to the imputation of insensibility. The same expedient was resorted to by M. Angelo, in the figure of Abijam, and by Raphael in the expulsion from paradise.

These were the artists who formed the second school of art, and established its end and limits. On it was founded the third period of style, in which refinement induced a grace and beauty not to be surpassed. The masters of this period were—Appelles, Protogenes, Aristides, Euphranor, Pausius, and the pupils of Pamphilus, and his master Eupompus. The last named artist was of Sicyon; and his authority was so great, that out of the Asiatic and Grecian chords of painting he formed a third, by dividing the last into the attic and Sicyonian. Pliny says that, when consulted by Zysippus on a standard of imitation in art, he pointed to the crowd passing by—observing that nature, not an artist, should be the object of imitation. Pamphilus, a Macedonian, the master of Appelles, adopted the doctrines of Eupompus. To the art of painting, he joined the study of mathematics, and held, that, without the aid of geometry, no artist could arrive at any perfection. In Appelles, we are told by Pliny, unrivalled excellence was found. Grace was his powerful and peculiar faculty, in

which he surpassed all his predecessors. His Venus Anadyomene, which was long after purchased by Augustus for one hundred talents, or £20,000 sterling, was esteemed the most faultless creation of the Grecian pencil, the most perfect example of that simple yet unapproachable grace of expression, of symmetry of form, and exquisite finish, in which may be summed up the distinctive beauties of his genius. Hotogenes was next to him in merit. The most celebrated of his works was his figure of Jalysus, with his dog, which occupied him seven years. Aristides, of Theles, and cotemporary of Appelles, was the first, who, by the rules of art, obtained a perfect knowledge of expressing the passions and affections of the mind. In one figure, he expressed the anguish of maternal affection, and the pangs of death. Euphranor, the Isthmian, and pupil of Aristides, is said to have carried this refinement of expression still further. Skilled in Sculpture, as well as in Painting, his conceptions were noble and elevated; his style masculine and bold; and he was the first who distinguished himself, by imparting majesty to his heroes.

Asclepiodorus, the Athenian sculptor, as well as painter, was as the latter, celebrated for the beauties of a correct style, and the truth of his proportions. Appelles allowed himself to be, in these respects, as much his inferior to this artist as he was to Amphion on the good ordering and disposition of his figures. About this period appeared Nichomachus, Nichophanes, Pyreicas and others. Nichias, an Athenian, 322 B. C., was in great repute for the great variety and noble choice of his subjects, for the mode of distributing his lights and shadows, and for great skill in the representation of animals. In Rome, 300 B. C., Fabius, a noble Roman, painted the Temple of Health, and gloried so much in the art, that he assumed the name of Pictor. Without a further enumeration of masters, for a long period after the reigns of Vespasian and his son Titus, painting, as well as sculpture, continued to flourish in Italy. Even under their successors Domitian, Nerva and Trojan, they met with as much encouragement as they did in the most palmy days of Greece, although painting was used in mere ornamental decorations—that is, to a great extent, and artists had sunk, for the most part, to the standard of mere mechanics. Under Adrian, Antonine, Alexander Severus, Constantine, and Valentinian, the art of painting continued to be an object of interest; but at length, in the reign of Phocus, with the fall of the Empire, with the rest of the noble arts and sciences, it was involved in the common heap of ruins.

(To be continued.)



BROMINE.

We give the folk wing interesting combination and experiments with bromine, as found in Hill's Treatise:—

"Discovered in 1826, by M. Balard, a young chemist of Montpellier, France. He named it *Muride*, because obtained from the sea; but it acquired its present name from a Greek word, signifying *rank*, or *dead odor*." It exists in nature in very small quantities. It is found in the waters of every sea which has been tested for it: it is also found in many mineral and salt springs. We have not been able to determine to whom belongs the honor of discovering its use as an accelerator in Daguerreotype. Having corresponded with Prof. Morse, (who was certainly one of the first who took portraits by the Daguerreian process,) Dr. Draper, and others, to whom this honor has been awarded, they refuse to claim it. Prof. Silliman, who is supposed to know everything relating to science, writes to me that he does not know this. I am quite inclined to place the wreath on the brow of the inventor of the Magnetic Telegraph.

PROCESS.—When common salt is prepared from sea-water by evaporation and crystallization, a liquid remains which goes under the name of *bittern*. This liquid on passing chlorine through it, is tinged with a deep yellow color. The liquid is now distilled, and the vapor passed over a substance, (muriate of lime), which has a powerful attraction for water. A small quantity of bromine is thus obtained.—Another method is to agitate the chlorinated bittern with a portion of sulphuric ether. The ether dissolves the bromine, from which it receives a beautiful red tint, and on standing, rises to the surface. Agitate this solution with caustic potash, and the bromide of potassium and bromate of potassa will be formed. Evaporate the liquor and the bromide of potassium will be left from which bromine may be distilled.

PROPERTIES.—At common temperatures, it rapidly volatilizes, giving red vapors of a most disagreeable smell. Its color, when held between the eye and the light is a deep hyacinth red. Like oxygen, chlorine, and iodine, it is a non-conductor of electricity, and a negative electric. It boils at 116.5° , and congeals at 40° Fahr. into a brittle solid. It is a powerful poison; even its vapor would no doubt prove fatal, if inhaled in large quantities. A single drop placed in the beak of a bird destroys it instantly. Operators cannot be too cautious in using it. A very small drop spattered in the eye would destroy the sight. Bromine is very corrosive. A lighted taper burns for a few moments in its vapor, with a flame green at its base, and red at the top, and is then extinguished. It is soluble in water, alcohol, and ether; the latter is the best solvent. With water at 32° Fahr., it forms a *hydrate*, in crystals of a fine red color. It

gives to a solution of starch an orange color. Chlorine will displace it from all its combinations with hydrogen.

CHLORIDE OF BROMINE.—Formed by transmitting a current of chlorine through bromine, and condensing the disengaged vapors by a freezing mixture. The factitious article is more simply formed, and is equally good as an accelerator, but not as quick. See page 25, Part I. M. Bissou, a Frenchman, found that the real chloride of bromine is so sensitive, that Daguerreotype proofs are taken by it in half a second. He succeeded in taking persons and animals in the act of walking.

BROMIC ACID may be obtained by pouring sulphuric acid upon a dilute solution of bromide of baryta, and evaporating. No interest.

BROMIDE OF BARYTA.—Boil of protobromide of iron with moist carbonate of baryta; carefully evaporate and it will crystallize in white rhombic prisms, which have a bitter taste, are slightly deliquescent, and soluble in water and alcohol.

BROMIDE OF CARBON.—Formed by mixing one part of periodide of carbon with two of bromine. Two compounds are formed, the *bromide of carbon*, and the *sub-bromide of iodine*; the latter is removed by a solution of caustic potassa. It is liquid at common temperatures, but crystallizes at 32° Fahr.; sweet to the taste, and of a penetrating ethereal odor; distinguished from the protiodide by the vapor which it emits on being heated. The *periodide of carbon* is made by mixing an alcoholic solution of pure potash and of iodine. It forms crystals of a pearly lustre, sweet to the taste, and of a saffron odor. The *protiodide* is formed by distilling a mixture of the preceding compound with corrosive sublimate. Sweet in taste, and of a penetrating ethereal odor.

BROMIDE OF MAGNESIUM.—Dissolve magnesia in hydrobromic acid. It will crystallize in small acicular crystals, of a sharp taste, very deliquescent and soluble.

BROMIDE OF SULPHUR.—Pour bromine on sublimed sulphur. There is formed an oily liquid of a reddish tint.

BROMIDE OF PHOSPHORUS, is formed by bringing phosphorus and bromine into contact in a jar filled with carbonic acid gas. Vaporizes by heat, and is decomposed by water.

BROMIDE OF SILICON, is prepared by burning silicon in the vapor of bromine. A very dense, colorless liquid, emitting dense fumes. We have used this article as an accelerator, and it produces a beautiful but singular picture. It works quick, but possesses no particular advantage.

BROMIDE OF ZINC.—Prepared by digesting a solution of bromine with zinc filings. The iodide is formed in a similar manner. No interest.

HYDROBROMIC ACID.—Mix the vapor of bromine with about an equal bulk of hydrogen gas, and introduce a coil of red-hot platinum wire. Red-hot iron answers equally well. The combination takes place slowly without explosion. Or, it may be formed by placing a small piece of phosphorus in a glass tube filled with water, and dropping it upon a little bromine. The hydrobromic acid passes over in the form of a gas, and may be passed through water, which will absorb it. *Hydriodic Acid* is formed in the same way, using iodine in place of the bromine. Hydrobromic acid is decomposed instantly by chlorine and nitric acid.

The French and German bromine is generally considered the best; but the American manufacture is by no means to be rejected, as it is frequently very excellent. Bromine is sometimes adulterated with naphtha.



NITRATE OF SILVER.

DECOMPOSED BY THE CURRENT OF A GALVANIC BATTERY.

Mr. Matteucci has observed that the black deposit obtained on the cathode, in the electrolysis of this salt, instantaneously becomes white when the current ceases; that it does not occur except when the solution is weak, because from strong solutions, crystalline silver is at once deposited; that if a portion of the deposit, after it has become white, be suspended between the electrodes while the current passes, those portions towards the anode again become *black*, if the experiment is made in a solution of the nitrate, but not if in mere acid water; it will, however, occur if a mere drop of nitrate is added to the solution, if a plate of a glass is interposed between the electrodes, the phenomenon does not occur; this is attributed to the great reduction of intensity. It is suggested "that the black deposit is formed, of oxide of silver, which is preserved by the passage of the current, and which, when the current ceases, passes immediately to the metallic condition."



OXIDE OF SILVER.

Place pieces of silver in a glass vessel,, and pour on them about equal parts of water and strong nitric acid; the metal will soon dissolve, giving off fumes of nitric oxide. Should the solution have a green hue, which is invariably the case, unless the metal has been obtained fine from the refiners, it indicates the presence of copper, in which case immerse some pieces of copper in the solution, and the nitric acid, by elective affinity, will combine with the copper; and a precipitate of pure silver, in the form of grayish powder, will take place. Throw away the liquid and wash the silver precipitate several times in sulphuric acid and water, and afterwards in water alone. Then re-dissolve it as before, in nitric acid and water; and a *solution of pure nitrate of silver* will be obtained. Place this in an evaporating dish, or a saucer, and apply the heat of a spirit-lamp, or place the saucer by the fire-side, till some portion of the liquid is driven off in vapor. Allow the residue to cool, and it will shoot out into long colorless transparent crystals, which are *nitrate of silver*. They must be handled with care, as they possess the property of staining animal and vegetable substances with an almost indelible black;—fused nitrate of silver being the lunar caustic of surgery,

and the main ingredient of marking-ink. Next prepare some *lime-water*, by stirring lime into water and filtering the solution. As lime is very sparingly soluble in water, requiring, at 60° Fahrenheit, 750 times its weight, it is necessary to make an abundant supply. Place the lime-water in a glass or other vessel, and drop in it a few crystals of nitrate of silver; the colorless solution will instantly assume an unsightly brown hue; and, after remaining quiescent for a time, the *oxide of silver* will subside in the form of a dark brown precipitate. The liquid is then poured off, and the precipitate is washed with water. Before throwing away the liquid, fresh lime-water should be added to it; and if the dark hue recurs, the precipitate must be allowed to subside again; if no change takes place, it may be inferred that the silver is all extracted. The oxide of silver should not be dried, but be kept in bottles with water.



POTASSIUM.

CYANIDE OR SIMPLE CYANURET OF POTASSIUM.

Before entering into the preparation of this ingredient, I may mention that the *prussiate of potash* of commerce has, by many young experimentalists, been mistaken for the above article, because it is often sold, by those unacquainted with chemical technicalities, under the name of *cyanuret of potassium*. It is, in chemical parlance, termed *ferro-cyanuret*, from its containing a certain portion of iron; and differs in its properties, very materially, from the *simple* salt. It is of a bright yellow color, and is converted into the colorless, simple cyanuret in the following manner:

Take 4 ounces of the yellow prussiate, break it in small pieces, and well dry it on a plate of iron; then reduce it in a mortar to exceedingly fine powder. Dry and pound in like manner one and a-half oz. of carbonate of potash. Incorporate the two ingredients thoroughly. Place a Hessian crucible in the fire; and when it attains a red heat, throw into it the prepared mixture, and closely cover the crucible. Keep up the heat, and the contents of the crucible will soon fuse; and the fluid mass will become red-hot. After this, immerse in it, from time to time, a hot glass rod; the mass which adheres, is in the early stages of the process brown on cooling; as the heat is continued, it appears yellowish, and finally, colorless and transparent. The operation is then complete: the crucible must be removed; and after its contents have been allowed to settle, the fused mass may be poured off; the greater portion of which consists of the *simple cyanuret of potassium*. This salt is very deliquescent, and must therefore be retained in close bottles: it will readily be recognized by its powerful odor,—that noticed in peach blossoms. The mere mention of prussic acid entering largely into its composition, will be sufficient to induce my reader to exercise common caution in handling it.

Having these ingredients prepared, take one pint of pure rain or distilled water; add to it two ounces of the cyanuret of potassium, shake them together occasionally, until the latter is entirely dissolved; and allow the liquid to become clear. Then add a quarter of an ounce of oxide of silver, which will very speedily dissolve; the dissolution may be hastened by heat, and after a short time, a clear transparent solution will be obtained.—*Walker*.

STEREO-CHROMIC PAINTING.

A new mode of fresco painting, called *Stereo-chromie*, which has for a long time excited attention among the artists of Germany, has it appears been perfected by a pupil of Kaulbach, in his own studio in Munich. The design is Kaulbach's, and the work was executed under his superintendence. It represents the figure and character of a Prophet, and will be sent to the London Exhibition. 'The figure, grand and majestic as it really is, cannot be expected to convey to the English public anything but a faint idea of the genius and skill of Kaulbach,' but as a specimen of this new and wonderful mode of painting, it will be regarded, by artists, with the greatest curiosity.

Stereo-chromie was discovered by *Obergrath von Fuchs*, a distinguished chemist; and is considered by German artists as one of the greatest discoveries of the age. They claim for it great advantage over ordinary fresco and encaustic painting, in its superior durability and the power which it affords the artist of retouching and glazing his picture. The colors are mixed with water, the whole being permanently fixed by occasional sprinklings of water, in which a certain proportion of fluoric-acid (*Flassspath saurs*) is mixed.

Stereo-chromie is in fact a preserver of the wall on which it is painted, By a certain chemical action of the solution sprinkled over the picture, while in progress, the whole ground on which it is placed, and the picture itself becomes one hard flinty mass, the very colors of which are converted into the hardest stone.

This singular species of painting resists, it is said, every influence of climate; and may be securely used as an external coating for buildings in any part of the globe. Neither is the artist confined to any particular time in executing it, leaving off when he pleases and for any length of time. In these points it is a most important recommendation, which cannot by any means apply to fresco work, nor, except within certain limits, to oil painting. The highest advantage of all, however, is that the same part may be painted over as often as the artist please, which is impossible in fresco; and consequently the most perfect harmony may, by this new mode be preserved throughout the largest possible painting. In fresco the artist is the slave of his materials; here, he is the arbitrary master and to the fullest extent.—*Banner of the Union, Pa.*

WATER.

Water intended to aid in conducting chemical experiments, should, so far as possible, be free from all foreign substances. As the Daguerreotype process is one of the most difficult in the range of chemical science, the Daguerreian should have at hand the best that circumstances will admit. We have learned, from experience, the disadvantages attendant on those who travel in the country, from village to village in the practice of the Daguerreotype art.

When *Distilled Water* can be obtained, it should be used. It must have no smell, taste, or color; it must evaporate without leaving a residue; it must give no precipitates with solutions of nitrate of silver, chloride of barium, superoxalate of potash, caustic alkalies, carbonated alkalies, sulphurated hydrogen, or hydrosulphate of ammonia.

Rain Water generally contains carbonic acid, carbonate of lime, and chloride of calcium.

Spring Water contains the same impurities as rain water, superadded to chloride of sodium, sulphate of lime, carbonate of iron, &c.

River Water.—More free from salts than spring water, but often contains a large quantity of decomposing vegetable and animal matter. This is particularly the case with water taken from the Hudson, from Troy to its mouth; also, the Mississippi, and, in fact, almost all large streams which are navigated.

Lake Water.—All descriptions of stagnant water abound in half-putrified organic remains, as all well know, that the smaller the bulk of liquid, as in the case of the swamp or marsh, the more impure and the more unwholesome it is.

To purify Water.—Much has been said about purifying water for Daguerreotype use. We look upon this, so far as chemical agents are employed, in most instances a loss of both time and money; let every one *filter* the water for use *often* and freely and we will guarantee that *success* will be his reward. We will class water in the following order:—

1, Distilled water; 2, rain water; 3, brook water; 4, well water; 5, spring water; 6, river water: yet the water from some of our Western lakes would, we think be fully equal to rain water. We have used the water from Seneca lake, and have found it possessing less animal and vegetable matter, than found in the water from any well which we have been compelled to use.



ISINGLASS, as it is generally called fish glue, is prepared from the air-bladders of the sturgeon. It is found in commerce under various forms, such as large strings, small strings, and leaves. It is always white, semi-transparent, dry, fibrous, horny, of a faint odor and insipid taste. When a very thin leaf is moved between the eye and the light of the sun, a species of chatoyance is perceived. If macerated in cold water, Isinglass swells and softens; if boiled in water, it dissolves almost without any residue and forms a solution, which, on becoming cool produces a semi-transparent jelly. There are several counterfeits, but all are easily detected.



PAPIER-MACHE.

PAPIER-MACHE.—Pulped paper moulded into forms. It possesses great strength and is a valuable article in the manufacturing of cases for Daguerreotypes. It may be rendered partially water-proof by the addition of sulphate of iron, quicklime, and glue, or white of eggs, to the pulp; and incumbustible by the addition of borax and phosphate of soda. The papier-mache Daguerreotype cases, tea-trays, waiters, snuff-boxes, &c., are prepared by pasting or glueing sheets of paper together, and submitting them to powerful pressure, by which the composition acquires the hardness of a board when dry. Such articles are often inlaid with mother of pearl, and japanned, and are then perfectly water-proof.



CHLORIDE OF CALCIUM.

CHLORIDE OF CALCIUM.—This is so easily manufactured that any Daguerreotypist can make it with but little difficulty. Saturate dilated muriatic acid, say three pints of water to one of acid; this should be filtered through a sponge and then allowed to stand until all the aqueous solution has evaporated and the calcium crystallized. This may be improved by it being placed in a crucible; and fused with a quick fire, then poured out on a flat stone. When cold, it should be broken and put in close bottles. This is much used by Daguerreotypists as it possesses a strong affinity for water; consequently, it is a valuable auxiliary in keeping the iodine dry. It is also much used for drying gases and absorbing water from ethereal and oily liquids, in organic analyses.



MURIATE OF AMMONIA.

MURIATE OF AMMONIA—*sal-ammoniac*.—This is used by *few* to bleach Daguerreotype impressions, and relieve them of the "*blues*;" should, however, the plates be *well* cleaned and the mercury at a proper temperature, there will be no need of this preparation: yet, as a number have requested, we give the following combination, knowing that all Daguerreotypists are troubled more or less with solarized impressions, they may be more particularly with the *blues*. Make a saturated solution of muriate of ammonia, in pure water, and filter through paper. This reduced with an equal quantity of water before used; when the linen or any portion of the impression is badly solarized, after removing the coating with the hyposulphite solution, thoroughly rinse the plate with water, then pour the bleaching solution over the surface of the impression in the same manner as in gilding.

If the solarization be very deep, apply the lamp beneath and slightly warm the plate, pour suddenly off, and without rinsing, quickly apply the gilding and gild in the usual way. The whole operation must be quickly performed, or the chlorine soon attracts the shade of the picture. There may be instances when this solution may prove of advantage, as, for instance, when black velvet and milk white are wanted in the same impression. We have seen it operate with pleasing success; but repeat that *few* only use it, yet as in a single instance it might be of value, we have given it. As much of the muriate of ammonia, in common, is adulterated, we will give a few tests which all can try.

When pure, this salt is totally volatilizable by heat; if a small portion on being heated on a piece of platinum foil over a spirit lamp leaves any fixed residue, it is adulterated. It should dissolve entirely in water; if it leaves an *insoluble residue* it is adulterated. The impurities generally found in *sal-ammoniac*, are sulphate of ammonia, sulphate of soda, chloride of sodium, and chloride of potassium; neither of these are considered injurious in the *bleaching solution*. Occasionally it contains lead, iron, and copper. The presence of sulphuric acid may be detected by means of a diluted solution of *chloride* of barium, added to a weak solution of muriate of ammonia, in water, this will produce a white precipitate, which is the more copious and dense, the less the liquor is diluted. If it contains lead add to a small portion of it, contained in a test tube, a little diluted *sulphuric acid*, or a few drops of a solution of *sulphate of soda* (glauber's salt), if the liquid contains lead there will be prevalent a white powder, or precipitate, this powder scarcely dissolves at all in diluted acids, but it dissolves in a solution of caustic of potash: *iron*, mix a solution of the yellow prussiate of potash with a solution of red prussiate, a few drops of this added to a

weak solution of muriate of ammonia will produce a blue precipitate; *copper*, pour about half an ounce of the liquid in a test tube, and add to it a few drops of liquid ammonia, if copper is present, the liquid will assume a blue color. Sal-ammonia is known by giving out the odor of ammonia when mixed with caustic of potash: when sal-ammonia possesses a *brownish* color, it indicates the presence of charcoal, or empyreumatic oil; such sal-ammoniac is good for some purposes, but wholly unfit for *chemical purposes*.

When sal-ammoniac leaves a non-volatile residue, it may contain *sulphate of soda*. This is the principal cause of failures with the bleaching solution, as the sulphate of soda has a tendency to blacken rather than bleach Daguerreian impressions. The sulphate of soda as well as the chloride of sodium, is often found in unpurified sal-ammoniac to the amount of ten per cent.

When sal-ammoniac contains much sulphate of ammonia, it fuses and sputters before it sublimes; whereas it otherwise sublimes without fusing. When sal-ammoniac, sublimes without residue, but gives a precipitate when its solution is tested with a solution of chloride of barium, it contains sulphate of ammonia; but when it leaves a non-volatile residue, the precipitate indicates sulphate of soda or sulphate of magnesia.



BROMIDE OF SILVER.

NATIVE BROMIDE OF SILVER AND ANALYSES.

M. Berthier says, that in the district of Plateros, seventeen leagues from Zacatecas in Mexico, silver ore is found in two different states; first, native silver; and secondly and principally in a state of combination in small olive-green or yellowish crystals, supposed to be chloride, but which he found to be bromide of silver. According to M. Duport, from whom M. Berthier received these specimens, this substance is not rare in Mexico, but occurs frequently in fine cubic and octahedral crystals.

The specimen examined by M. Berthier was from San Onofe. It was compact, of a slightly reddish gray color; fracture uneven; splendent; penetrated with small cavities, some of which were partially filled with a substance of a dull pale yellow color, and which the miners call oxide of lead; other cavities contain very small imperfect crystals; which are brilliant, and of a pale olive green-color, and have the appearance of chloride of silver. This specimen was very rich, for it yielded 0.0688 of silver, and contained 0.45 of carbonate of lead, which, intimately mixed with quartz and a little oxide of iron, formed the principal portion of the mass.

M. Berthier has also found this mineral among the silver ores of Huelgoeth, department of Finistère in France. Two specimens were obtained by him: the first of these is described as being porous or scoriform, containing white quartz imbedded in foliated hydrate of iron. On the edges of the foliated iron ore the naked eye could distinguish small cubic grains of a pearl-white color, which had all the characters of chloride of silver.

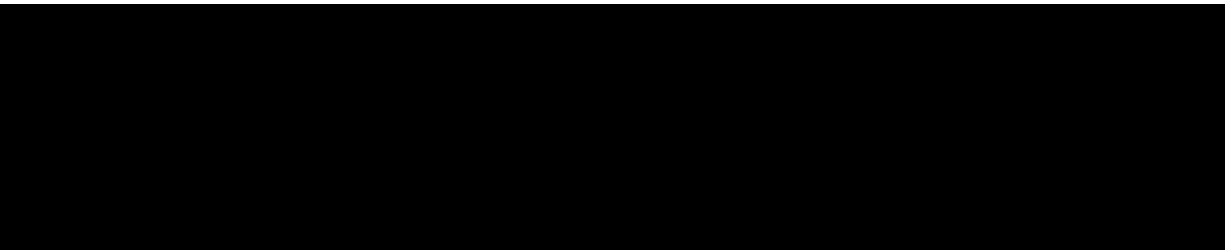
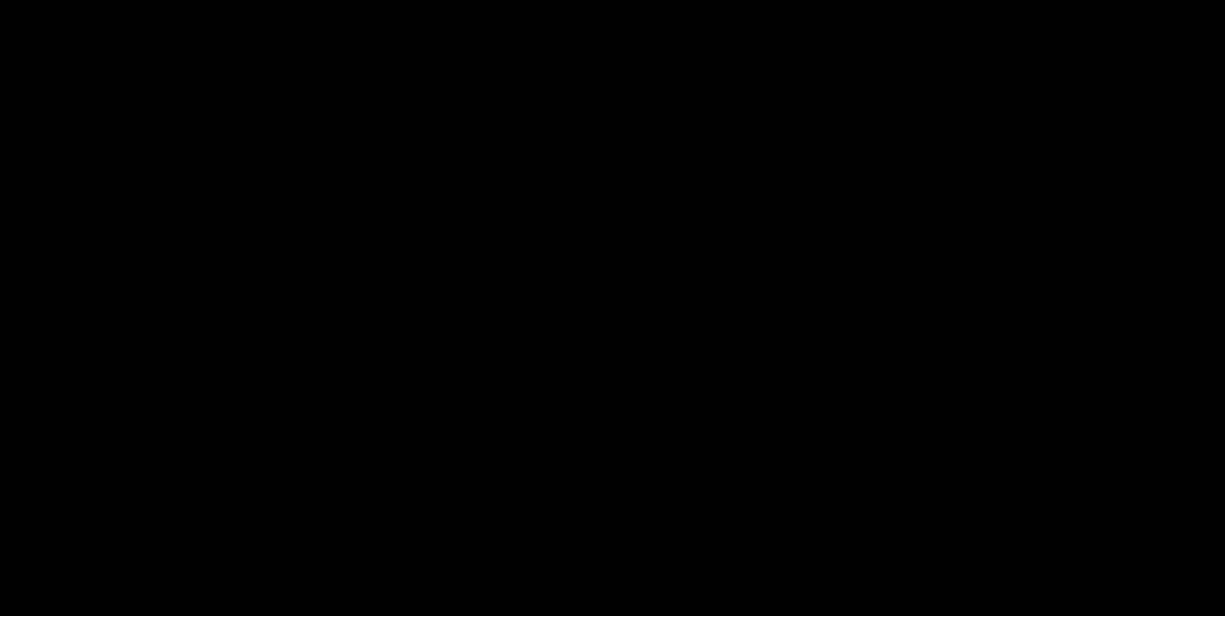
The second specimen had the appearance of compact oxide of iron, containing here and there milk-white quartz; it was throughout impregnated with chloride of silver, which occasionally appeared in the form of very small brilliant crystals. To analyze this mineral, 10 grammes were first treated with ammonia, and heat to dissolve the chloride of silver, and afterwards by boiling hydrochloric acid to dissolve the oxide of iron; this acid also dissolved a certain portion of lead, which probably was in the state of phosphate. The quartzose residue weighed 32.6 grammes: it contained 0.17 gramme of silver, which must have been in the metallic state: the ammoniacal solution gave by boiling and saturation with nitric acid, 1.84 gramme of chloride of silver, which, supposing it to be pure, contained 1.40 gramme of silver, which, added to 0.17 gramme remaining in the quartz, gives a total of 1.57

gramme; a result which differed so very little from that obtained by assaying, as to prove the absence of bromide of silver, and that this was the case was confirmed by additional experiments.

After this a third specimen was received from Huelgoeth; it was very small, but as rich as the foregoing, and in it there were distinguishable, besides granular cubic crystals of chloride of silver, other grains of an olive-green color, which had exactly the same appearance as the bromide had. Plateros and the following experiments proved the presence of the substance.

Five grammes of the pulverized mineral were boiled in a solution of oxalic acid, until the oxide of iron was perfectly dissolved: the residue weighed about a gramme, and it evidently contained a mixture of canary-yellow and white grains. It was digested in hot solution of ammonia until all the yellow powder disappeared; it required a large quantity of the alkali for this purpose, which would not have been the case to dissolve pure chloride. The solution was gradually saturated with nitric acid, and it was observed that the successive deposits formed had an evident yellow tint, but gradually diminishing in intensity, except the last, which were white. The yellow deposits were collected and examined in the following manner:—A portion was treated with chlorine and æther; the æther became of a yellow color. Another portion was dissolved in ammonia, hydrosulphate of ammonia was added to the solution, and the black precipitate formed was separated, and was found to be pure sulphuret of silver. The liquor was concentrated by exposure to the air, and filtered to separate the sulphur which was deposited; a little potash was then added, and it was evaporated to dryness; acetic acid was added to saturate the excess of potash, and it was again dried.

To determine whether the saline residue contained a bromide, a small portion of it was treated in a tube with pure nitric acid, and a yellow liquid was immediately obtained. Another portion was mixed with peroxide of manganese, and the mixture was placed in a glass tube; a few drops of concentrated sulphuric acid were added, and when gently heated, red vapors were immediately disengaged, and after some time there were deposited on the sides of the tube small drops of a red liquid. The existence of bromine was therefore evident, and it was proved that the bromide was unmixed with iodide. Bromide of silver appears to be rare at Huelgoeth; but it may be readily distinguished from the chloride by its greenish or canary-yellow color, which is characteristic of it. It is remarkable that it occurs with the chloride in the same specimens, but without there being an intimate mixture of the two substances.



The Daguerreian Journal.

NEW YORK, MARCH 1, 1851.

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HILLOTYPÉ.

Since the announcement of the HILLOTYPÉ, in the last number of the Daguerreian Journal, quite an excitement has prevailed, both in the Daguerreotype and scientific world. The great question has fairly been solved and "*natural colors*" can be produced and rendered indelible upon the metallic plate. America can safely say she has presented to the world one of the most invaluable discoveries that has ever been imprinted upon the pages of history.

Men profound in their scientific skill and learning, have long and in vain sought for the discovery or invention of some means of securing to the future, the colors of the present. All Europe has been alive to this great desideratum, and many have presumed it an impossibility, while some few persisted, flattered by hope and encouraged by the almost daily announcement of some new discovery or invention, that heretofore had been deemed only as having existence in the dreamy imagination or a perspective future. We hail the discovery of the Hillotype as an epoch bright in the history of science, as well as impregnated with interest to aid in unfolding a volume of investigation which has so long commanded the attention of learned and philosophical minds, as the "*colored rays of light*."

The subject of reflected light and its colors has long agitated the scientific mind. Much has been said and written; elaborate and laborious "Essays," "Researches" and "Treatises" have penetrated our libraries and proved valuable auxiliaries in pushing the interest of those engaged in furthering agricultural and chemical pursuits. This branch of natural science is regarded as one of the most important, and, at the present moment, it is rendered doubly interesting from the fact of Mr. Hill's discovery. We may look forward for new developments, which will prove no less surprising than that of rendering permanent, on a metallic plate, the variegated beauties of the solar spectrum.

We are, as it were, standing upon an eminence from which we can survey the present, retrospect the past, and almost sketch the bright outline of a coming future. Here lies a field for animated speculation, in which nature's student can satiate his appetite in the study of nature, in a province hitherto unknown and unexplored; he may here realize truths purely sublime, painted in the glowing "*colors of nature*," and rendered prominent upon the tablet of his memory.

We present the following communication from Mr. HILL, which our readers will peruse with pleasure:

For the Daguerreian Journal.

S. D. HUMPHREY, ESQ.;—Being detained here a day on my way from your city, I relieve myself a little from the tedium of delay, by a few scribblings to your valuable Journal. I will give a few particulars respecting my pictures, &c.

I have now fifty-five specimens. They are all equally perfect. It is quite remarkable that I have never yet made a partial failure. Those impressions which have had too much light, are nearly as strong, sound, brilliant and beautiful as those correctly timed in the camera, being inferior only in having the colors less deep. Even the whites retain their strength. The folds of the linen are always well defined. Blue or solarized linen is unknown in my process, and there is always a strength and clearness in the whites, unattainable by mercury. During the past winter I have several times taken a view in which there is a deep red house, while the ground was covered with snow. For experiment I exposed the plate so long as to reduce the bright red of the house to a *very light red*, while at the same time, the white snow was developed with a beautiful whiteness.

I have copied several very highly colored French prints. The copies are far superior to the originals, in that, while they have *every* tint of color, they are exceedingly brilliant. This is a characteristic in which I never fail, even with the plates merely cleaned with rotten stone—the brilliancy depending on *other causes*. Well polished plates, however, are preferable for other reasons. It is *essential* that the plates should be very pure, free from scum? dampness, and organic matter of every kind, and I am experimenting with different substances, in hopes of finding something that will more perfectly cleanse, while it thoroughly polishes. I would be very thankful to any person who might furnish me with valuable hints on this point, as I am convinced that here lies one great cause of uncertainty.

My trouble with the *yellow*, which you mentioned in the last number of the "Daguerreian Journal," relates only to the homogeneous rays, orange, buff, and all the various shades of yellow come out true except the *chrome yellow* which appears less brilliant. This, however, is thought by distinguished artists, no serious objection.

My late visit to your city was much shorter than I could have wished. I called on a number of Artists, and the cordial manner in which they congratulated me has added much to my kindly feelings towards the fraternity, and strengthened my resolution to give all worthy Daguerreotypists and Artists, my process on terms which I believe will be satisfactory. As far as this is concerned, please say to such their interests are safe in my hands. I met with but one person anyway sceptical, and he is willing to be more fully satisfied when he "sees the pictures," which is very fair I am sure. While it is very cheering to me, in my truly arduous duties, to review the kind congratulations of my brother Artists, the most I am entitled to *claim* in an age

when almost every announcement is regarded as humbug, is that the Daguerreian world will give me a fair chance to perfect my discovery. A few have seen my pictures, and their expressed opinions agree with mine, viz., that these pictures will astonish the world. I saw in your city, in the hands of Mr. Hite, Artist, some exquisite ivory miniatures. It astonished me that the human hand could paint "the form of the human face divine," in such a mode; but I pleasantly said to the Artist, what I now repeat in all seriousness, that the pictures by my process necessarily exceed in beauty the finest productions of the painter, they being drawn by light and painted by sunbeams.

I have heard several rumors while in the city, in respect to my designs in the disposition of my discovery. Allow me to say, for the information of all concerned, that my purpose is *unalterably* fixed to avoid *monopoly*, and to take a course which will put the process into general use.

To-morrow I start for my "mountain home," and with my invigorated health and spirit, I hope to resume my toils with a prospect of bringing out my process at an early day.

I remain,
fraternally yours,
L. L. HILL.

WOODSTOCK, ULSTER CO., N. Y.,
March 12, 1851.

✍ We are highly gratified to find many of our fellow Daguerreotypists are cordially welcoming us on by manfully and numerously sending us their names for the Daguerreian Journal for one year. When we commenced our editorial labors we launched our hark upon a sea never before thought safe for regular navigation, but we find only few snags in the way, and these are every day lessened by our close application and the assistance of kind friends. No man knows his friends until they have an opportunity to prove themselves.

With but few exceptions, all who subscribed for the Journal for the first four months have sent us two dollars worth of the "root of all evil," for their subscription for the balance of the year. We *bow*, and *thank you!* May you never get in a "Fog."



✍ T. ANTISELL, M. D., has been appointed Professor of Chemistry in the Vermont Medical College, at Woodstock, Vt. He is now fulfilling his professional duties at the above named institution.



☞ Mr. J. E. MAYALL of London, has promised us an article on "Photographing on Glass." This will appear as soon as received.

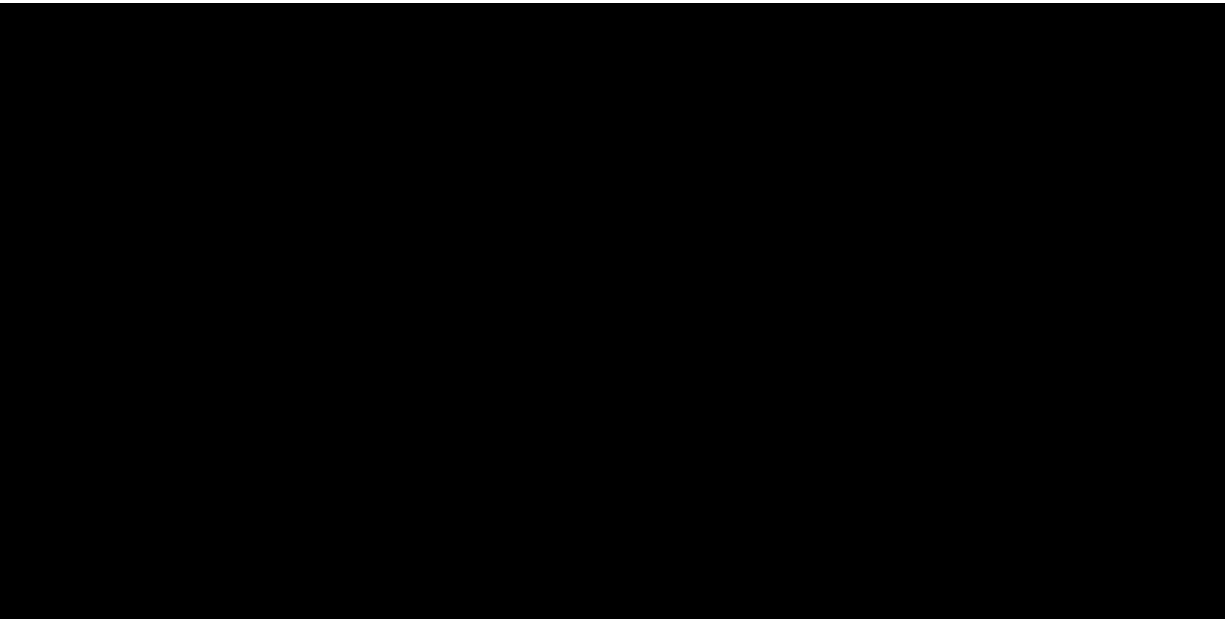
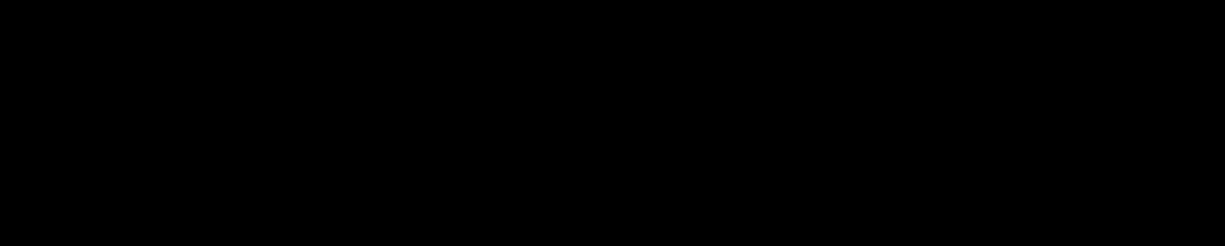


☞ Our old and learned friend Mr. FINLEY of Canandaigua, promised us a communication. Where is it?



☞ REMOVAL.—In our last we noticed our removal to No. 252 Broadway; this was the case, but we found on consulting a carpenter in relation to putting in a sky-light, he declared it unsafe. The consequence was, that we at once abandoned the idea, and soon made very satisfactory arrangements with Mr. INSLEY, the well and favorably known proprietor of the *Insley Gallery*. Once again we are settled and would be happy to see all who feel an interest in the Daguerreian and Photogenic arts.

The Office of the Daguerreian Journal is at No. 311 Broadway.



Our Daguerreotypes.

[REDACTED]

[REDACTED]

We are assured by one in whom we have all confidence, and withal! capable of judging, that there is now in progress a machine for cleaning and buffing plates. This machine is an ingenious and valuable contrivance, and it is said will polish a plate in one-sixth of the time required by any other process. The whole done without handling the plate. *We hope so.*



GURNEY has recently taken some of the finest large size Daguerreotypes ever produced. These wonderful specimens are on plates eleven by thirteen inches, called mammoth plates. Such pictures four or five years ago would have filled the world with admiration and surprise. The chemical effect is clear and well worked, thus proving well prepared plates can present large as well as small impressions. It is well worth while for every Daguerreian visiting this city, to look upon these proud specimens of the art.



THOMPSON of this city, has just taken the whole of the upper floor of the building occupied by him, and has built two fine sky-lights.



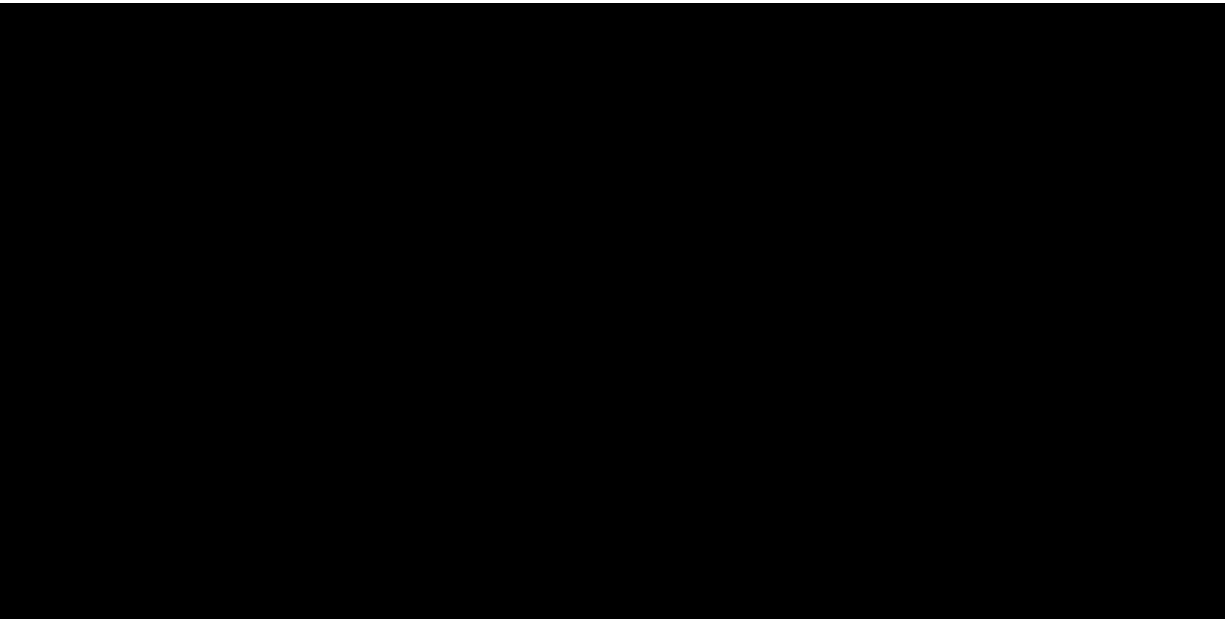
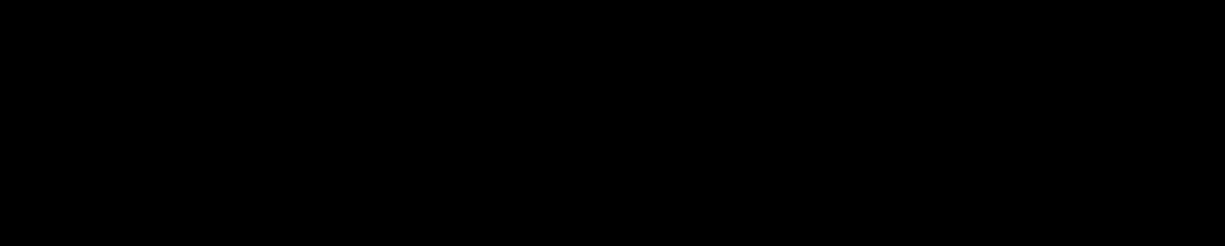
We understand that Mr. H. McBRIDE, operator for MEADE & BROTHER of this city, is about to establish himself in Albany. We wish Mr. B. success.



WESTON of this city, is now producing some of the best Calotypes we have seen. We are happy to see Photographing on paper brought before the public. Daguerreotypists should call and see Mr. W's. specimens, as well also those produced by Messrs. Bertha, Wehnest, Beekman & Brothers, who have been for some time past successfully practising in the art of "Sun Drawing."



A. MORAND has recently made very extensive additions to his former establishment in Chatham street. We now find him lighting his subjects by a large and well arranged sky-light.



Correspondence.

S. D. HUMPHREY, ESQ.,

DEAR SIR:—I would fain give my testimony in favor of your excellent Journal, which is to the Daguerreian, as a guide to the traveller, and pilot to the untaught mariner. It needs but to meet his eye for him to appreciate its value and secure its aid, while following in his misty pathway.

I have long since learned that "experience is the best teacher," but now since the advent of the Journal, all who would be relieved of difficulties incident to Daguerreian life, may find the cause of *atmospheric troubles—bad light—poor subjects—clouded results, &c.*, reflected on its pages, and thus, by close application, prevent the many evils which so frequently attend them, and thereby learn that "an ounce of prevention is better than a pound of cure."

But there are those in the business who do not seem to discover any difference between the profile on the school-boy's slate, and the best "aqua-tinta" engraving. *They* never have any trouble in producing *perfect* results at *every* sitting, and *that* for the sum of one dollar or fifty cents. No wonder the writer of the article on Daguerreotypes in No. 5 of the Journal, says "*Daguerreians* are not *artists*, nor Daguerreotypes works of *Art*, but *mere wonders of Nature*."

When Daguerreians adopt the motto—"Good pictures and fair prices," they may lay claim to the enviable name of Artist, and not before. The Daguerreian art is the most difficult art to practise with *perfect* success, that has ever yet been discovered. A *perfect* Daguerreotype is the result of a series of the most careful, delicate and complicated experiments ever conceived of by the human mind.

I have been considerably amused in perusing the Journal, in reading communications from sources where I am acquainted. They remind me of the old adage—"All is not gold that glitters," for they appear to far better advantage in print, than as *Artists*. But enough of this. In conclusion, permit me to speak of a method of copying pictures, which, if new to any of your readers, they will find it well worthy of the trial. I find, by placing the picture to be copied where the rays of the sun may fall obliquely upon the plate, all reflection is thrown off, and the image appears in the camera, clear and distinct, even if the original be very faint. It also shortens the time of exposure to from one to five seconds.

Respectfully yours,
JAS. BENNETT SYKES.

OWEGO, March 4, 1851.



NOTE.—We think this is rather sharp firing, but, as it is already met by the "Artists" referred to, we give it. We do like *spice*, and well seasoned articles, but don't get too personal. A man may have been born in a *stable*, but it is no *sign he is a horse*. We hope the above from Mr. S. was not prompted by "a spirit of rivalry and animosity," but rather for the good of all, as no doubt it will so prove. Give us a *pop* from the other side. [ED.]



DAGUERREIAN JOURNALS LOST.—The following Telegraph Dispatch was received at the Daguerreian Journal Office on the 13th.

"What is the reason of your not sending me my Journals? W. S GEAR."

The following was our answer forwarded per mail on the same day.

DAGUERREIAN JOURNAL OFFICE,
NEW YORK, March 13th, 1851.

DEAR SIR:—Your Telegraph Dispatch came to hand. In answer,—I do send a Journal directed to you as often as published. The last was mailed last Friday, i.e. the February 15th number, that being the last out. I find it takes nearly as many to furnish the Post Offices as my subscribers; however, I try it again and send one to-day.

Respectfully,
S. D. HUMPHREY.

W. S. GEAR.

We wish that the persons, wishing the Daguerreian Journal would furnish us with their names—"no questions asked") and they can have a copy forwarded,—thereby our *honest* subscribers would receive their Journal regularly. We do know that we put the Journal in the Post Office—but we don't know *why* so many of our subscribers do not receive them. There is no doubt but that our Journal is valuable, and should be in the hands of all, yet it is with us as with most brother editors, we are unable to furnish copies gratuitously.



AMMONIA.

Ammonia should be a perfectly colorless liquid; when concentrated its specific gravity should be 0·9. It must have a pure ammonial odor, and must volatilize without any residue. It must not become troubled when mixed with alcohol, or lime water, otherwise it contains carbonate of ammonia. Neither must it for the same reason, become troubled when tested with a solution of chloride of barium; this is the best test for carbonate of ammonia. Care is to be taken to observe the action of the test on its first addition, for after the mixture has been exposed to the air for a few minutes, a troubling necessarily ensues, because the ammonia abstracts carbonic acid from the atmosphere.

When super-saturated with nitric acid, and tested with a solution of *nitrate of silver*, it must give no white precipitate; otherwise it contains sal-ammoniac. It must give no precipitate with a solution of chloride of barium, otherwise it contains sulphuric acid, in which case, the precipitate will be invisible in nitric acid. It must give no precipitate with a solution of oxalic acid or superoxalate of potash; otherwise it contains lime. When saturated with nitric acid and tested with a solution of yellow prussiate of potash, it must give no precipitate; otherwise it contains copper. When mixed with its weight of oil of vitriol, it must not become brown or black; otherwise it contains empyreumatic oil. As it is quite difficult to obtain ammonia entirely free from carbonate of ammonia, or to preserve it in that state, the employment of ammonia containing a small portion of the carbonate is admissible in the practice of the Daguerreotype.



AN ARMY OF SUBSCRIBERS.—It is computed that the regular subscribers to the *New York Sun*, standing side by side, and each occupying eighteen inches space, would form a line *seventeen miles, fourteen rods, and three yards long*. Standing with arms extended, and occupying six feet each, they would reach *sixty-eight miles, fifty-eight rods and one yard, &c. &c.—Sun*.

Let us see if we also cannot get up some astonishing statistics: The regular subscribers to the *New York Tribune*, standing in single file, each holding fast to the other's coat-tail, would form a line 176 miles, 7 rods and one yard long. Allowing that each could hop, skip and jump the distance of 4 yards, their simultaneous performance of this feat would extend from Cape Horn to Baffins Bay. Supposing they all sneezed at once, the vibration of the air would overthrow the spire of Trinity Church. In reading the 48 columns of our double sheet every morning, the distance travelled by their eyes would reach twice round the earth. There! Is that enough?—*Tribune*.

Wonder if the Tribune's calculations is made from the actual measurement of the "tail" of the "old white coat?" If so its not fair, for Horace has the advantage over the modern fashion.



STRANGE ANNOUNCEMENT.—We see by a Georgia exchange, that some one in that State is taking "Daguerreotypes as low as seventy-five cents." Our Devil says "Barnum is after the *wonder*, and intends having him early this Spring."



EXAMPLES OF THE DIVISIBILITY OF MATTER.

All that has been written in support of what is termed the "Infinite Divisibility of Matter," has been advanced with an oversight of certain axiomatic principles which immediately decide the question in the negative. Indeed, it would suffice to say that there can be no such thing as a division of matter into an infinite number of parts simply because there can be no such thing as an infinite number of any thing. Number is essentially finite, and although, in imagination, it can be extended indefinitely, it never can be positively infinite, at any period of futurity, from the very fact that we can always conceive of its being extended still farther. Even as an ideal extension of parts, it has always a beginning and an end, at any given time we may assume for measuring it, and is, therefore, never infinite in itself at any time. In this respect it differs essentially from infinite space, which having no parts, is positively infinite at all times. Number is a convenient instrument wherewith to obtain an idea of infinite space, or of infinite duration, on account of its capacity for endless extension or continuation; and it is because we can ideally extend and continue it forever, without even approximating to a measurement of space and duration, that we discover the latter to be absolutely infinite, and number to be absolutely finite. We discover that between the capacity for endless extension and endless extension itself, there is a manifest and decided difference. Under any circumstances, therefore, the term infinite, as applied to the divisibility of matter, is unphilosophical and improper. All the matter in the universe is finite, and if it were to be doubled, or quadrilliontupled, every instant of time, to all eternity, it would never become infinitely extended. And, by the same rule, it can never become infinitely subdivided.

If, however, by the term "infinite divisibility," as applied to matter, is merely meant its capacity for interminable subdivision, then the question resolves itself either into one of speculative fancy or of practical fact. If the former, then it must be admitted that, by an effort of the imagination, we can conceive of such a divisibility beyond any assignable limits. But if we regard the question as a practical one, it immediately becomes too absurd for serious attention, since it is evident that human power, limited in all things, must be so in producing artificial subdivisions of matter. However unlimited, therefore, the capacity of matter may be for divisibility, that of human beings, in relation to it, must be narrow indeed. When chemists, therefore, and other philosophers, speak of the divisibility of the present ultimate atoms of matter, they must, however unconsciously, regard them in relation to human ability, and thus far the question may become one of experiment. But the question as to whether matter can actually be subdivided indefinitely and without end, is one readily answered in the negative, by the known inability of mankind to continue an

endless experiment of this kind. The whole question therefore, concerning the illimitable divisibility of matter, which has been discussed from the earliest period of science to the present day, is frivolous, fruitless, and irrational.

As matter now exists, not only its ultimate atoms, but even its constituent molecules, are as inconceivably minute, as its aggregations are vast. Taking an exemplification, from organic matter, we find that a single drop of a strong solution of indigo, in which at least 500,000 parts are rendered distinctly visible by the microscope, colors 1,000 cubic inches of water, and as this quantity of water is at least half a million of times greater than the drop of indigo solution, the particles of indigo must be smaller than 2,500,000,000,000, the twenty-five hundred millionth part of a cubic inch. If we dissolve a particle of silver, of 0.01 of a cubic *line*, in size, in nitric acid, it will render distinctly milky 500 cubic inches of a clear solution of common salt; and, consequently, the magnitude of each particle of silver thus divided and diffused, must be somewhat less than the billionth part of a cubic *line*! So great a number as a billion being but imperfectly conceivable, it may render the idea of this minute division more distinct, by stating that to count a single billion of seconds, by a watch, every day and night without ceasing, would require 31,675 years. In gilding silver wire, it is found, by calculation, that a grain of gold is spread over 1,400 square inches; and as, when examined by a microscope, the gold upon the thousandth part of a linear inch is distinctly visible, it is demonstrated that gold may be divided into particles of at least the billionth, 400 millionth, of a square inch in size, and retain the color and all other characteristics of a California prize. If a grain of copper be dissolved in nitric acid, and then in water of ammonia, it will give a decidedly violet color to 392 cubic inches of water; and, therefore, if there was but one particle of copper in each portion of the water of the size of a grain of sand, of which one million make a cubic inch, it would show that the original grain of copper had become divided into 392 particles.

This extreme tenuity of matter, however, is far surpassed in some examples which may be adduced from organic sources. It is upon authentic record that an Irish girl has spun linen yarn, of which one pound was 1,432 English miles in length, and of which, therefore, 17 pounds and 13 ounces would have girt this world; and yet less than the 127 millionth part of this thread would have been distinctly visible, and must have contained other filaments still finer, each of which must have been composed of an indefinite number of smaller particles, themselves, in all probability of complex organization, and containing certainly, minuter atoms of carbon, besides those of gaseous matter.

If we employ the microscope, far greater wonders of divisibility than these appear, even in the complex organization of animal life. Ehrenberg has shown that tripoli, a mineral much used in the arts, is entirely composed of the siliceous shells of the microscopic animalcula, known as infusorials, and that a single cubic inch of it

contains at least *forty-one thousand millions* of these shells!—about fifty times as many individuals as there are of human beings on the globe. Yet each of these minute animals lived and fed; had digestive and circulative systems, with blood possessing globules as large, probably, in proportion, as those of ours, besides nerves of sensation and inclination, with brains belonging to them, together with muscles and every other mechanical apparatus for the extremely active locomotion and propagation which they so interestingly exhibit. And every new improvement in the microscope reveals new races of animals, apparently created for their own enjoyment, and of which millions heaped on millions, would be utterly invisible to the unaided eye.—These facts prove that things are great and small only by comparison with each other, and not in relation to infinity, with which no comparison can be instituted. And they prove, moreover, that vain, boastful, and bigoted man, is not the sole object of creation, nor of the profound benevolence, contrivance and design, with which it is universally replete.—*Art's Echo*.



IRON—ITS NATURAL AND ARTIFICIAL COMBINATIONS WITH CARBON.

The question is often asked, what constitutes the difference between wrought iron, cast iron and steel?

Cast Iron, when viewed under favorable circumstances, by the help of a microscope will be found to be a mechanical aggregation of molecules of *iron and carbon*; and the relative position of these particles may be illustrated by a pile of cannon balls as usually arranged in navy yards, each alternate ball being iron and carbon (charcoal).

If a mass of cast iron be heated until softened, and then *puddled* (squeezed), the carbon will be forced to the surface, and will there combine with the oxygen of the atmosphere, forming carbonic acid or carbonic oxide gases, and thus pass off. When all the carbon has been parted with, the mass is called Wrought Iron, and may then be welded, when at proper heat, but cannot be melted—*the hottest blast furnace will not melt wrought iron*. Wrought iron at red heat combines rapidly with oxygen, and becomes oxide of iron—thus a joint of stove-pipe thrown into a furnace will never melt, but by contact with atmosphere will change into oxide of iron, and thus be practically lost. This operation is technically called *burning*. If a piece of wrought iron be surrounded by carbon (charcoal) finely pulverized, and the whole enclosed in a sheet iron vessel to exclude the air, and this placed for a sufficient length of time in a furnace constructed for the purpose, the iron will imbibe an atomic quantity of carbon, and become Steel. This process is called *cementation*, and steel so made can be melted as readily as cast iron.

Thus it will be seen that both cast iron and steel are combinations of iron and carbon, and in the same proportions, but not in the same state of combination. In cast iron, the carbon and iron are a mere *mechanical combination*, while in steel the iron and carbon are combined *chemically*.—Wrought iron, when pure, is free from carbon, and its ductility, toughness, &c., are due to the absence of carbon, sulphur, phosphorus, and other substances, with which it is occasionally pervaded.

The French chemists are experimenting, and occasionally succeeding by accident, in causing heated iron to take the carbon from carbonic acid and other gases containing carbon, and thus becoming steel more rapidly, and at less cost, than when made by the process of cementation. Mr. Dixon, of Jersey City, has succeeded in making steel direct from the Adirondack iron ore, while Peter Cooper, Esq., Mr. Dickinson, and others, are manufacturing wrought iron direct from the iron ores of

New-Jersey without first forming the pig or cast iron, and of course at less expense, as the saving of fuel is very great.

The process of *case-hardening*, or changing the immediate surface of iron utensils into steel, is readily performed by covering their surfaces with such organic substances as contain carbon as a constituent, and then subjecting them to high heat for short spaces of time—thus the roller of a paper or sugar mill may be case-hardened by a coating of prussiate of potash, or of leather chips, and then subject the whole to high heat, excluded from atmospheric influences. By this process the gelatine and other constituents of the leather are reduced to carbon, and this enters the surface particles of the iron, converting them into steel. Many hypotheses are offered for this action, and among others, that "*the ultimate particles of matter are always in motion*," admitting the ingress of particles travelling in smaller orbits between them. The friends of this hypothesis offer as proof, that a fresh cast sash-weight when broken is a gray mass, while one taken from an old building, and broken, is beautifully crystalline, from the centre to the outside, like speculum metal. A freshly drawn piece of tin pipe when suddenly bent opposite the ear gives no crackling sound, and if broken has no crystalline structure, but if left at rest for one hour it has both. Barbers often tell us that razors get tired of shaving, but if laid by for thirty days they will then shave well. By microscopic examination it is found that the *tired razor*, from long stropping by the same hand and in the same directions, has the ultimate particles or fibres of its surface or edge all arranged in one direction, like the edge of a piece of cut velvet; but after a month's rest, these fibres re-arrange themselves heterogeneously, crossing each other and presenting a saw-like edge, each fibre supporting its fellow, and hence cutting the beard, instead of being forced down flat without cutting, as when laid by. These and many other instances are offered by the friends of the hypothesis named, to prove that the ultimate particles of matter are always in motion, and they say that in the process of welding, the absolute momentum of the hammer causes an entanglement of orbits of motion, and hence a re-arrangement, as in one piece; indeed, in the cold state, a leaf of gold laid on a polished surface of steel, and stricken smartly with a hammer, will have its particles forced into the steel so as to permanently gild it at the point of contact.

The oxidation of metals is equally curious, and the length of time necessary for the formation of an infinitesimal coating of oxide is less than the one-thousandth of a second. This fact may be readily proved: a clean surface of steel, free from oxide, when brought in contact with mercury (quicksilver) will amalgamate, but if the least oxide be upon the surface no such effect will take place. Prepare a trough containing quicksilver, and place a bar of steel above it, and within one inch or less of the surface of the quicksilver—break this bar with a smart blow from a hammer, so that the blow which breaks it shall at the same time force the broken ends into the quicksilver, and although the time occupied by the ends in passing through one inch

of atmosphere before reaching the surface of the quicksilver will be immeasurably short, still they will be so oxidized as not to amalgamate with the mercury; if, however, the bar of steel be confined at its ends below the surface of the quicksilver, and then be broken *upwards*, by a lever applied to its centre, the ends of the broken bar will be beautifully amalgamated before reaching the atmosphere above. The reason for the success of the last named experiment is doubtless due to the absence of oxide of iron, when broken beneath the surface of the mercury. J. J. MAPES.



NEW WEIGH-LOCK AT ALBANY.—A writer in the *Courier and Enquirer* gives a description of the great weigh-lock that has just been erected at Albany, for the purpose of ascertaining the tonnage of canal boats.

Heretofore long and vexatious delays have been the result, while now, a boat is brought into the lock, and in a time which would have scarcely sufficed in other days to have ascertained the weight of a small parcel, the unerring register on the beam, has registered its weight, and the record is on the books of the office. The weigh-lock is directly on the side of, and attached to the large canal, and is adapted to the use of such boats as shall hereafter be built, when the locks throughout the entire length of the canal shall be of the uniform enlarged size.



DAGUERREOTYPING IN LONDON.—In a recent letter from J. E. Mayall, dated London, February 11th, he writes in his postscript:—"While I write this letter we have the gas lighted in the streets—1 o'clock P.M., and I write by gas-light."

We must confess that the Daguerreotypists are in the "Fog" enough in this country, but if "Old Sol" should get blocked up at that early hour, we don't know what would become of our "Foggy" Daguerreotypists here.



The ground side of the "Ground Glass" should face the lenses.



Mercury baths should always be kept covered with some porous wood, in order that the mercurial vapors cannot escape in the room.



QUICK STUFFS.

We have endeavored to guard against giving useless receipts, thereby saving our Daguerreotypes a tax which would prove in most instances of no benefit. As many have requested us to publish a variety of receipts for Quicks. We give them with but little comment. We will number each combination:

No. 1. Take pure rain or distilled water, one quart filtered through paper into a bottle having a ground glass stopper, and add one and a half ounces of chloride of iodine for warm weather, or little less for cold weather. The reason of this is obvious, from the fact that during the warm summer day, the bromine is far more volatile than in a winter day. To the above add one ounce of best American bromine, and shake well. Now, with care, to prevent, as far as possible, the escape of gas, add, drop by drop, thirty drops of aqua ammonia, shaking well the mixture at each drop.

It is necessary that caution be observed, and not add more at a time than three drops of the aqua ammonia, as otherwise it evolves too much heat.

Use. Put in the box one part quick to eight parts water. Coat to dark yellow over dry iodine, and change to a deep rosed color over the Quick; recoat over iodine one-tenth.

No. 2. Lime water, one quart; chloride iodine, one ounce; add three-fourths of an ounce of bromine—shake well.

Use. Put in box one part quick to six water; coat to bright yellow over iodine; to rose over the quick, and recoat one-fourth.

No. 3. Take rain or distilled water, one quart; add pulverized alum until it is a little sour to the taste; and a small piece, say one half inch square, of magnesia,—Filter through paper, and add chloride of iodine, one half ounce; bromine sufficient to take it up, which is a little less than half an ounce.

Use. One part quick to six parts water; coat over iodine to a soft yellow, nearly, but not quite bordering on a rose; over the quick to a dark purple or steel; recoat from one-sixth to one-tenth.

The above works slow, but with a good light and proper management it can be made to produce as good impression as any combination known, yet it is not so easily managed as No. 1.

No. 4. Rain or distilled water, one gallon; bromine, one and a half drachm; sulphuric acid, two and a half drachms; hydrochloric acid, ^[C] two drachms.—Shake well as each is added in the above order.

^[C] This acid dissolves glass with great rapidity. It is purchased in leaden bottles. A single drop on the skin would make a sore difficult to heal. Daguerreotypists should bear in mind that the accelerating chemicals used in the Daguerreian art, are of the most volatile substance, and more difficult to experiment with than all other in the range of chemical science. When hydrofluoric acid is to be measured, the graduated glass should be partly filled with the mixture to which you intend adding it.

Use. To one ounce water, add from fifteen to thirty drops quick; coat over an incipient rose; over quick, nearly change the color to fair rose; recoat about one-third or one-fourth as long as at first coating. The coating box should be charged strong enough to change the plate in from *one* to *four seconds*. One advantage this mixture possesses, it will work in one-half the time required for any of the foregoing combinations. One very serious objection to its use is, it cannot, without great care, be made to work with certainty; and another objection is, it will not last long, as the box will require to be replenished after having coated from *eight* to *fourteen* plates. The older, however, this mixture is, the more certain in its operation. We have one gallon which has been made *two years*.

No. 5. *Dry Quick*. We have already given this preparation in a former number of this Journal.

No. 6. *Acidulated Quick*. Water, one pint: bromine, ten drops; chloride of iodine, forty drops; nitro-muriatic acid, one-fourth ounce; sulphuric acid, two drops.

Use. One part quick to ten parts water; coat over iodine to orange; over quick to rose red; recoat one-fourth.

There are thousands of different combinations and agents employed, and, after all, let every Daguerreian make up his mind to first *become acquainted* with some *good* combination of bromine and iodine, there will be less complaining of *Quicks*.



Money received since our last Number.

W. A.; M. R.; F. S. H.; J. W. O.; S. B. D.; H. O. N.; T. O.; J. H. F.; S. B. B.; M. P. B.; A. B.; J. H. V.; W. A. J.; J. E. M.; M. M.; H. H. L.; J. W. H.; S. N. R.; N. E. S.; R. B. A.; H. S. B.; D. C.; T. C. D.; C. H. G.—each \$2. G. & B.; D. McD.; S. P.; M. S. U.; C. T. M.; S. S.; W. R. R.; S. B. jr.; E. N. H.; C. W. T.; J. M.; S. H.; N. C.; C. M. H.; J. B.; J. B. R.; W. O. G.; N. P. S.; L. O.; A. T.; W. S. W.; N. E. R.; D. G. K.; L. Q. V.; H. K.; Y. M.; A. G.; U. B.; D. A. N.; V. T.; W. O. O. V.—each \$3. S. B. & CO.; M. & B.; P. C.; T. C. D.; S. & S.; H. & M.; M. W. N.; G. D. A.; J. S.; A. T. W.; M. A. H.—each \$5. N. E. S.; T. C. D.—each \$10. M. S. \$15. B. F. \$34.



ANSWERS TO CORRESPONDENTS.

W. O. R., Mo., Mr. R. writes—"Will you have the kindness to inform me of any new process for preparing phosphate of iron. I make the request more particularly to obtain a process by one Mr. Routh, who I understand has a new process."

The preparation referred to by Mr. R., was read before the London Medical Society, January 11, 1851, by Dr. Routh, at which time he exhibited specimens of phosphate of iron made by dissolving the ordinary phosphate in meta-phosphoric acid, and by then evaporating to degrees. It has been found that this form of iron is exceedingly useful in cases where iron is desired.

W. B. N., Mexico—Mr. N. says—"Will some of your subscribers give, through your Journal, a practical and first-rate process for producing Calotypes?"

We hope some of our "subscribers" will furnish the same.

H. B. T., Ohio.—Mr. T. says—"Will you give a ready and easy method of detecting chlorine? I find in a former number of your valuable Journal, that you have been making experiments with this gas. I have had remarkable success with this gas, although not a practical Daguerreotypist."

Mr. J. will find 'this gas' is readily distinguished from other gases by its color, odor, and bleaching properties. Probably the most simple method of detecting free chloride is to hold a rod dipped in aqua ammonia over it, when white fumes will be formed.

Will Mr. J. have the kindness to forward to us his experiments—we would like much to lay them before our readers.

A Stock Dealer, in one of the Southern States, writes—"What articles of stock will be less needed by artists, provided Mr. Hill's process proves true, of producing pictures with all the 'colors of nature?'"

Colors and brushes!

J. W. S , Mich.—We forwarded the article you ordered, and enclosed the balance in the package. The view camera you had better order soon, as it will be some time before Mr. Harrison can get it ready, he having so many orders now on hand. You had better have one of Lewis's *new* Camera boxes.

T. J. C., Va.—We cannot recommend the articles you speak of; they involve only an old vague principle, and are only worthy the person having them in charge.

A. G. L., Pa.—The money received and forwarded to Europe—your plan may work, but it looks a little too "opaque." You will find by heating an iron plate, and applying it to your apparatus, will facilitate the operation much. Chapman has the exclusive right to the patent.

E. W., Miss.—See page 24, Daguerreian Journal.

R. M. H., Ala.—You will find "Silliman's Journal" a great aid and valuable work, published the first day of every second month, price \$5 per year. New Haven, Conn.

See answer to L. I. G., page 187.

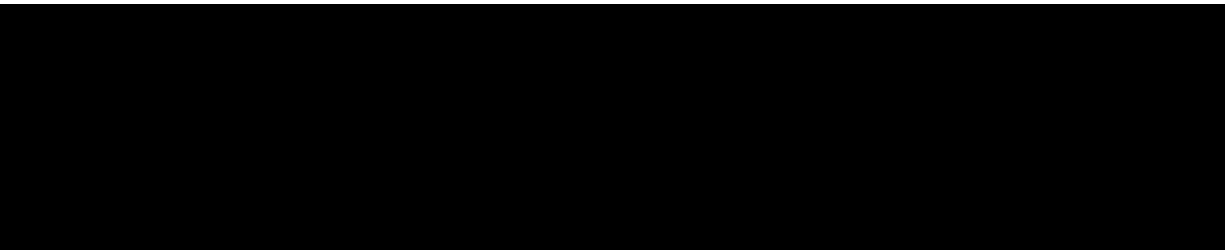
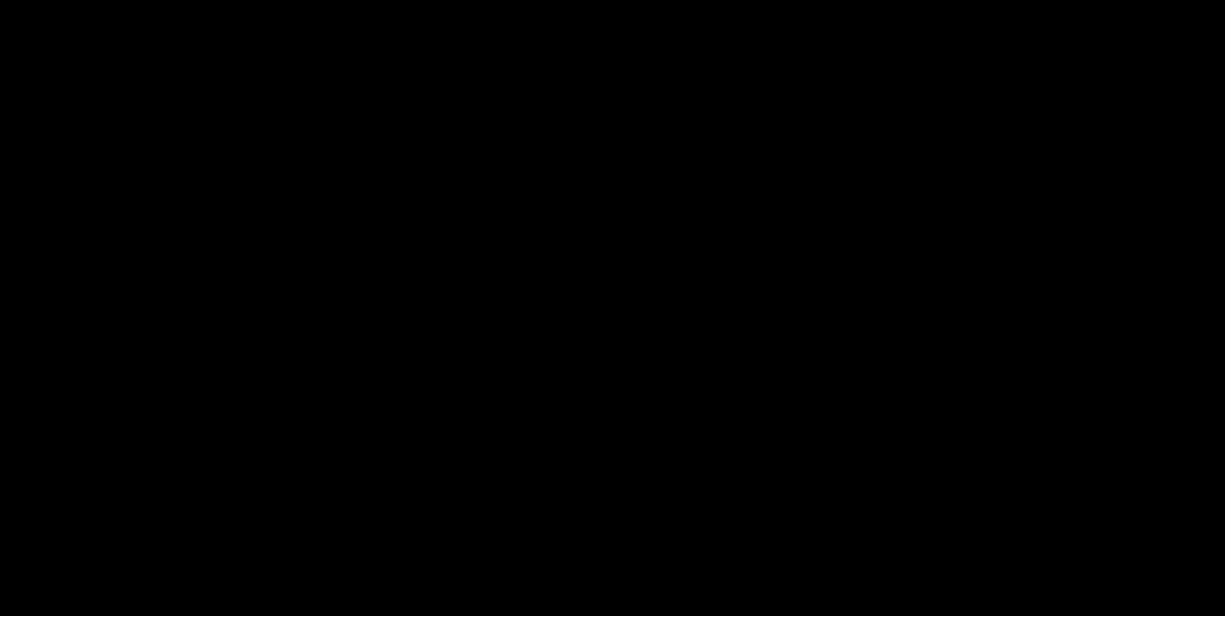


NOTICE OF NEW PUBLICATION.

The New York Register of Medicine and Pharmacy, published semi-monthly and edited by Dr. C. D. Griswold.

This valuable Medical Journal is fast making its way into the medical ranks with marked energy. Its age is about the same as ours, and we are happy to see that with us it is prosperous; every number is received and read with pleasure. We find the following pithy notice in the Register, which is only one of many from the cute pen of the Editor, who it seems has 'cut up' as well as received a "New Dido." "We were presented with the first number of the "New Dido" by the publisher, and from the title, at first supposed that it might be in some way connected with the medical profession, inasmuch as some 'pranks' in a medical way have been 'cut up' in poetry, but in looking it over, we find that a couple of 'chaps' have started off for a drive, although what they are 'driving at,' the author has not yet disclosed. We think it likely enough they may "turn out" students, for they are evidently after subjects, which no doubt will be brought to light in the next number."—Good, Dr., that is worth the Dollar for the subscription.

We have also received the 15th of March number, which has a fine likeness of Dr. John W. Francis.



ADVERTISEMENTS.

HILL'S TREATISE ON DAGUERREOTYPE.

TWO VOLS. IN ONE.

The whole Art made easy, *and all the recent improvements revealed*. Containing also—The Process for Galvanizing Plates, and the whole Art of Electrotpe; the Reproduction of Daguerre's Images by Tithonotype; an account of Calotype Paper, and other methods of Photogenic Drawing &c. By L. L. Hill, Westkill, Greene Co., N. Y.

CONTENTS.

History of Photography; Theory of the Process; Description of Apparatus; Account of Stock, such as Plates, Cases, Chemicals, and other articles to be purchased forming a complete Daguerreian Directory; Recipes, a large number; Polishing Plates; Coating the Plate; Camera; Mercury; Gilding; Coloring; Calotype, Tithonotype, &c.

The above is but a partial outline of the subjects treated in this work. Nothing is omitted which is necessary to render any person of the requisite judgment and taste "a workman that need not be ashamed," The *Recipes* have cost the author more than \$500, and no pains or expense have been spared to render the work a complete Manual of Daguerreotype.

TESTIMONIALS.

The following extracts from a large number of highly flattering Testimonials from distinguished Artists and others:

M. A. ROOT, an eminent Daguerreian, Philadelphia says, "I can freely say yours is, in my opinion, a most valuable Treatise on the Art of Daguerreotype."

From the *Albany Express*, "We would recommend this useful work to all practising the Art."

E. JACOBS, a celebrated Operator, New Orleans, says, "I have perused the work with much gratification as being much needed in the Art. I can sell a large number to

pupils."

From the *Ulster Telegraph*, Saugerties, N. Y.: "Those acquainted with the Christian character, and extraordinary skill of Mr. H. in Daguerreotyping, need no recommend from us. His pictures are perfectly exquisite, and his modes of operating of the most simple and scientific character. The Book will be a treasure to those engaged in the Art. It is one of those rare works which is as good as its title-page, and we heartily recommend it to the Daguerreian fraternity."

Orders should be directed to S. D. HUMPHREY, Agent, New York.

Also for sale, as above, price \$2, a pamphlet by the same author, on the Magic Buff, Plate-Making, Fixing Process, Celerotype, &c.

TO DAGUERREOTYPISTS,

WANTED—A PARTNER to engage in the Daguerreotype business. One having from Five Hundred to One Thousand Dollars can now have an unequalled opportunity to join with one of the best Artists in the country, the advertiser possessing unequalled advantages for opening a Room in this city, now offers a rare chance to any one possessing the above amount of capital, none other need apply. Applications to be made by letter, which will be *confidential*, and addressed, *post-paid*, to the Editor of the Daguerreian Journal, References must accompany the answer to this, and real name stated. A chance for Daguerreotypists out of the City.

PETER SMITH,

IMPORTER and Dealer in Tubes, Apparatus, Plates, Cases, Chemicals, and every article appertaining to the Daguerreian Art.

No. 36 Fifth Street, Cincinnati, Ohio.

LOUIS BECKER

WELL KNOWN CHEMICALS, for sale at BECKER & PIARDS, No. 201 Broadway, N. Y.

DAGUERREIAN ESTABLISHMENT.

JOHN ROACH, OPTICIAN, 79 Nassau Street, New York. Has always on hand

VOIGHTLANDER INSTRUMENTS, quarter, half, and whole size.

American Instruments, Roach's make, warranted to be superior to any yet made in the United States. They work with sharpness, and quickness, and persons purchasing can test them with the high priced German Instruments.

Coating Boxes, Flint Glass Jars, cemented in, and ground air tight.

Mercury Baths, with Thermometer Scale in front. Head Rests, Stands, Cases, Chemicals, &c. Plates, French 40th of the Star, and other first quality Brands.

Bromine Roach's Triple Compound, now called QUICK-STUFF, works with certainty and quickness, in all weather, and pictures taken with it have a rich white tone.

GALVANIC BATTERIES, &c.

DAGUERREOTYPE GOODS ONLY.

ANTHONY'S NATIONAL DAGUERREIAN DEPOT.

205 Broadway, N. Y.

THE attention of Daguerreotypists, and the Trade, is respectfully invited to my assortment, which I believe to be unequalled in extent and variety.

Plates, Frames, Case-
Cases, Cameras, Makers'
Chemicals, Apparatus, Materials,

of every style and size.

For Sale, price \$1 per copy, beautifully bound in cloth, the standard work, second edition, enlarged and greatly improved, the History and Practice of the Art of

Photography, or the production of Pictures through the agency of light, by H. H. SPELLING, illustrated with thirty-five engravings.

Goods can be forwarded to order, to any town or village in the United States or Canada, and the bill collected on delivery of the goods, provided such town has connection by express with New York. Where there is no such connection, Daguerreotypists would do well to order their goods to the nearest express town.

E. ANTHONY,

Importer and Manufacturer of Daguerreian Materials.

N.B.—Good journeymen Case-Makers wanted, to whom steady employment will be given.

BENJAMIN FRENCH,

No. 109 Washington Street, Boston.

DAGUERREOTYPE Apparatus, Plates, Cases, Frames, Gold Lockets, Polishing materials, Chemicals, and every description of Goods used in the Daguerreotype business, constantly on hand and for sale, at wholesale and retail, at the lowest cash prices.

3tf

S. J. THOMPSON,

WOULD most respectfully announce to the public, that he has one of the best sky-lights in the United States, and is prepared to execute Daguerreotypes. Likenesses of all sizes, put up in every style of the Art.

No. 57 State-street, Albany, N. Y.

2ly

\$5 REWARD.

STOLEN from the door of Clark Brothers, 551 Broadway, one full size Daguerreotype View, in papier mache frame, oval fire gilt mat. Said View of a GOTHIC COTTAGE, on the steps of which can be seen a lady, two or three boys and a dog. Any person returning the above described Picture, or giving information where it may be found, shall receive the above reward Oct 16.

WANTED IMMEDIATELY.

A GOOD DAGUERREOTYPE OPERATOR can have steady employment to engage in a pleasant western city in the state of Ohio.

Also a first class operator can have good wages and engage for one year, to go to California.

For further particulars enquire at this office. None need apply but such as are competent to take *full charge* of the operating department of a large establishment; salaries will be in accordance.

J. D. WELLS,

DAGUERREIAN ARTIST, No. 6 Kirkland's Block, Main Street Northampton, Mass. *Likenesses* taken by a sky-light connected with a beautiful side-light. Pictures put up in all styles of the Art. Plates, Cases, Locketts, Frames, and all kinds of Daguerreotype Stock for sale. 2-3

Circular to Daguerreotypists.

GEORGE DABBS & JAMES CREMER, Travelling Agents for L. Chapman, 102 William street, New York, manufacturer of Daguerreotype cases, mats, preservers, and plates, and importer of the *genuine* Star brand, No. 40, French plate, and last, though of first importance, proprietor of "Peck's patent plate holder,"—the great desideratum which only requires to be used to be appreciated. Prices, \$1,00 for medium; \$1,50 for quarters; \$2,00 for halves and \$2,50 for whole size holders—a vice to hold the blocks \$1,50 and an instrument for bending the plates 75 cents. They would inform Daguerreotypists and dealers that they will wait upon as many during

the winter, as their time will permit, for the purpose of exhibiting the patent Plate Holder, for a description of which see advertisement headed "Two New Inventions"

NEW YORK, November 1, 1850.

1tf

INSLEY'S DAGUERREOTYPE GALLERY

REMOVED TO
No. 311 BROADWAY, N. Y.
(*Between Stewart's and the City Hospital.*)

THIS, our new Sky-light Gallery, is located on the second floor at the above number, and is universally acknowledged to be the most convenient and effective Gallery in the City. Every real improvement is taken advantage of, and, aided by scientific and gentlemanly assistants, we trust our pictures cannot be excelled.

The Clergy—the Statesman—the Artist—the Man of science—and all lovers of really good Daguerreotypes, throughout the United States, are invited to call and examine our collection.

P. S.—This Gallery was for several years located on the corner of Cedar street, but is now removed to No. 311 Broadway.

J. E. MARTIN,

"EXCELSIOR ROOMS," Jefferson Avenue and Odd Fellows' Hall, Detroit.
Daguerreotype Likenesses taken in every style of the Art.

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CAMERAS.

C. C. HARRISON, Manufacturer of Cameras, and Camera Lenses, of all sizes and of the latest improvements. Office 293 Broadway, New York, where in his Gallery may be seen specimens of Daguerreotypes taken with instruments of his own

manufacture, which for accuracy of performance are unsurpassed by any in the world.

N. B. A new and improved Camera for taking views, is manufactured by him, at a price unprecedentedly low.

C. C. HARRISON,

No. 293 Broadway, N. Y.

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PREMIUM DAGUERREOTYPE DEPOT AND MANUFACTORY.

W. & W. H. LEWIS, 142 Chatham Street, New York, keep constantly on hand, superior CAMERAS of all sizes; also, *quick working* Cameras, fully equal to any imported. All kinds of Apparatus, including our Patent PLATE VICES and GILDING STANDS; Galvanic Batteries for Electrotyping, for durability superior to all others. Buffing Lathes, on the most approved plan, Plates, Cases, Chemicals, Polishing Materials of every description. All kinds of Walnut, Rosewood and Gilt Frames for Daguerreotypes, outside Show Frames, Diploma, Certificate and Picture Frames made to order. Pressing Machines, for Straw and Leghorn Hats, of improved construction.

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SCOVILL MANUFACTURING CO.

No. 57 Maiden Lane, New York,

Have constantly on hand an extensive assortment of all articles belonging to the Daguerreian Art; embracing plates of their own, and French manufacture, Mattings, Preservers, Frames, Cases, Lockets, Chemicals, Cameras and Apparatus of every variety.

Agents for the sale of C. C. HARRISON's celebrated Cameras.

All orders will receive prompt and careful attention.

CLARK BROTHERS, & B. L. HIGGINS.

Daguerreian Gallery, over the "Regulator," Franklin Buildings, Syracuse, N. Y., No. 128 Genesee St., Utica, Tremont Row Boston, and 551 Broadway, New York.

Likenesses by the improved Daguerreotype of various sizes, and of the most delicate executions may be obtained at the above rooms during the day, from 8 A M., to 5 P. M.

Chemicals, Plate, Cases, Cameras, Apparatus, and other materials, connected with the art, constantly on hand, and for sale at New York prices.

All articles are selected with great care and warranted to give the best satisfaction.

THE SUBSCRIBER, would respectfully inform the Daguerrian Artists, that he has *constantly on hand* a large assortment of Daguerreotype apparatus, plates, cases, and chemicals, comprising in part the following:

Voightlander & Sons, Harrison's, Lewis' and Perry's Cameras and other apparatus of the most approved styles.

PLATES—Scovill's, Chapman's and the different brands of French plates.

CASES—silk and velvet lined, Papier Mache and a great variety of fancy cases.

CHEMICALS—American, German and French Bromine, chloride of iodine, do gold, calcium, mercury rouge, rotten stone, black polish, colours, brushes, rosewood and Papier mache, frames, mats, glass preservers, prepared buck skin, &c., &c. Every article used in the business, which I will furnish to operators at retail or wholesale, as low as the same quality of articles can be bought in New York or elsewhere.

Our motto is small profits and quick sales.

N. E. SISSON.

No. 496 Broadway, Albany N. Y.

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TWO NEW INVENTIONS IN THE DAGUERRETYPE ART.

"PECK'S PATENT PLATE HOLDER," and the "*Bent Edge Daguerreotype Plate*," used in connection with it. An instrument is sold for seventy-five cents, with which every operator can bend his own plates. The holder is a desideratum, and only requires to be used to be appreciated. It is so constructed that it will hold the plate through all the stages of cleaning, buffing, polishing, coating, taking the picture in the camera, and mercurializing without any change. During the whole process, the plate need not be touched with the fingers, and does away with the use of wax, &c., &c.

The prices for the holders are mediums, \$1 00—quarters, \$1 50—Halves, \$2 00—whole size, \$2 50.

The "*Magic Back Ground*." The discovery of this is due Mr. C. J. ANTHONY, of Pittsburgh, Pa. By this process the most beautiful effects can be produced, and the pictures are set forth in bold relief on back grounds of various shapes and tints. Pictures taken with the "Magic Back Ground," will be emphatically the "*Pictures for the Million*." The Patent is applied for, and the right ratified upon the receipt of the Patent, for the sum of Twenty-Five Dollars.

L. CHAPMAN, 102 William St., N. Y.

Manufacturer of Cases, Mats, Preservers, Plates, Importer of the genuine Star Brand, No. 40 French Plates, and dealer in Daguerreotype stock generally.

LEVI CHAPMAN,

No. 102 William street, New-York, Manufacturer of, and Dealer in Daguerreotype Stock.

Daguerreotype Cases.

Medium size, from \$15 to \$198 per gross,

Quarter " " 24 to 288 "

Half " " 60 to 432 "

MATS, PRESERVERS and CHEMICALS of all kinds. French and American PLATES.

L. C. imports the genuine No. 40 Star Brand French Plate, which he keeps constantly on hand, together with an assortment of other kinds.

PAPIER MACHE CASES, inlaid with Mother of Pearl, exceeding in beauty any thing of the kind heretofore made, from 90 to \$1152 per gross.

L. C. is also proprietor of Peck's patent moveable Plate-holder.

GEORGE DABBS, }
JAMES CREMER, } Travelling Agents.

DAGUERREOTYPE FURNISHING ROOMS.

WM. A. WISONG.

No. 2 N. Liberty Street, Baltimore, Md.

HAS CONSTANTLY ON HAND, a full and general assortment of Stock, for Daguerreotype use.

Embracing every variety of Cameras, Plates, Cases, Fixtures, Trays, Chemicals, and Materials used by Daguerreian Artists, all of which are offered at the lowest market rates.

Orders from Artists are respectfully solicited, and forwarded with dispatch.

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DAGUERREOTYPE PLATES.

2000 LOUIS L. BISHOP'S superior PLATES, offered for sale at a great bargain, by

VICTOR BISHOP, 23 Maiden Lane. N. B. These Plates are silvered by the electro-magnetic process, and warranted superior to the best French plates.

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THE SUBSCRIBER still continues to carry on the business of ENGRAVING ON WOOD, in all its branches. His facilities are such that he is enabled to execute all orders promptly, and in every style of the Art, upon the most reasonable terms; while the experience of many years enables him to feel confidence in his efforts to give satisfaction to all who may favor him with their patronage.

Particular attention paid to the Drawing and Engraving of Subjects from Daguerreotypes.

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WILLIAM S. DORR, 101 NASSAU STREET, over Ackerman & Miller's Sign and Banner Painting Establishment, is prepared to print, in the best style, and at the usual *Low Prices*, Books, Periodicals, Newspapers, Pamphlets, Bill Heads, Circulars, Commercial and Law Blanks, Bills of Lading, Bank Checks, Notices, Labels, &c. CARDS printed on the celebrated Yankee Press.

Few offices in the city have greater facilities for doing work with *neatness* and *despatch*, as most of the materials are new, and long editions are done by steam power presses.

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Likenesses taken equally well in all weather.



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Baker, F. S., Baltimore, Md.
Broadbent, Samuel, Wilmington, Md.
Barnes, C., Mobile, Ala.
Bartlet, No. 102½ Main street, Boston, Mass.
Bogardus, A., No. 217 Greenwich street, N. Y.
Brown, F. A., Museum Building, Mashetnec, N. H.
Brown, H. S., Milwaukie, Wis.
Buxton, John, St. Catharine street, Montreal, Canada.
Bradlee, J. E., Boston, Mass.
Buell, E. M., Pittsfield, Mass.
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Bisbee, A., Dayton, Ohio.
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Brown, James, 181 Broadway, N. Y.

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Franklin Building, Syracuse, New York, and Tremont Row, Boston, Mass.
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Ennis, T. J., 106 Chestnut street, Philadelphia, Pa.

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Gay, C. H., New London, Ct.
Geer & Benedict, Syracuse, N, Y

Hill, R. H., Kingston, Ulster Co., N. Y.
Haas, Philip, No. 280 Broadway, N. Y.
Hall, W. H., Rouse's Point, Clinton Co., N. Y.
Harrison, C. C., 293 Broadway, N. Y.
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Johnson, Charles E., Cleavland, Ohio.

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Johnston, D. B., Utica, N. Y.

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Long, E., St. Louis, Mo.

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Stansbury, B., Brooklyn, L. I.
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Shorb, J. R., Winnsboro, S. C.
Shew, Myron, Chestnut Street, Philadelphia, Pa.

Thompson, S. J., No. 57 State Street, Albany New York.
Tomlinson, William A., Troy, New York.

Van Alsten, A., Worcester, Mass.
Vail, J. H., New Brunswick, N. J.
Van Loan & Co., 118 Chestnut street, Philadelphia, Pa.

Westcott, C. P., Watertown, Jefferson Co., N. Y.
Wood, R. L., Macon, Ga.
Whipple, John A., Washington Street, Boston, Mass.
Whitehurst, J. H., New York, Richmond, Norfolk, Petersburg, and
Lynchburg, Va., Baltimore, Md., and Washington City, D. C.
Wells, J. D., No. 6, Kirkland Block, Main street Northampton,
Mass.
Walker, S. L., Broadway, Albany and Poughkeepsie, N. Y.
Walker & Horton, Newburgh, N. Y.
Wentworth. Henry, Fort Plain, Montgomery Co., N. Y.
Williams, J. A., Washington Square, Newport, R. I.
Williams, J. B., Philadelphia, Pa.

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